



A survey on deep learning-based automated essay scoring and feedback generation

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Abstract

Deep learning-based automated essay scoring (AES) models exhibit a remarkable ability to identify complex patterns within essays and then generate accurate score predictions in an end-to-end training fashion. However, these models face a critical limitation in explaining the specific patterns and features utilized for scoring, which are essential for interpreting the scores and offering constructive feedback to essay authors. Numerous studies have focused on essay scoring, with the aim of modeling prompt-specific, domain-adaptable, or trait-specific AES. While existing surveys on AES cover topics ranging from representation to scoring models, they primarily emphasize scoring models. This study addresses a crucial gap by encompassing research on feedback generation for essay assessment tasks. By delving into essay scoring and feedback generation, we synthesize several existing literature to provide readers with a comprehensive understanding of ongoing research in both deep learning-based essay scoring and automated feedback generation. We categorized the existing essay scoring studies into prompt-specific and cross-prompt AES models, noting that prompt-specific AES is extensively researched category. However, we have only come across a few studies concerning automated feedback generation, likely because of the limited availability of suitable datasets for researching such types of tasks. Moreover, this survey provides insights into approaches for essay representation, prevalent datasets, evaluation metrics, and challenges in automated essay scoring tasks. By shedding light on these aspects, our goal is to delineate the current landscape, identify key research directions, and pave the way for further advancements in automated essay assessment.

Keywords Automated essay scoring · Automated feedback generation · Deep learning · Natural language processing

1 Introduction

Automated essay scoring (AES) systems represent computerized tools adept at evaluating essays and allocating scores. By harnessing text processing, natural language processing, and machine learning algorithms to assess essay quality, these systems streamline the grad-

Extended author information available on the last page of the article

ing process, offering benefits to both writers and evaluators. Through expediting grading procedures, providing prompt feedback, and mitigating grading inconsistencies, AES systems play a pivotal role in enhancing the efficiency and effectiveness of essay evaluation tasks (Williamson et al. 1999; Smolentzov 2013).

AES aims to replicate the role of human evaluators but approaches the scoring process differently. Human evaluators follow rubric guidelines where a score and its associated essay quality attributes are defined. For example, in the ASAP-AES dataset (see Sect. 6.1), essays are graded based on quality attributes such as content, organization, style, fluency and vocabulary usage (Hamner et al. 2012). In contrast, deep learning-based AES relies on the capability of neural network models to identify complex patterns in essays. These models demonstrate the ability to predict scores aligned with those of human graders. This has been showcased through various approaches, including simple deep learning models (Taghipour and Ng 2016; Alikaniotis et al. 2016; Dong and Zhang 2016; Dong et al. 2017; Cozma et al. 2018; Liang et al. 2018), trait-based neural models (Tay et al. 2018; Nadeem et al. 2019; Hussein et al. 2020; Song et al. 2020a; Kumar et al. 2021; He et al. 2022), transformer-based models (Yang et al. 2020; Cao et al. 2020; Wang et al. 2022b; Rodriguez et al. 2019; Han et al. 2023b; Xue et al. 2021), and cross-domain AES, which is particularly beneficial in scenarios with limited data in the target domain (Cozma et al. 2018; Dong and Zhang 2016; Jin et al. 2018; Cao et al. 2020; Ridley et al. 2020, 2021; Chen and Li 2023; Jiang et al. 2023; Do et al. 2023).

Despite the existence of papers on essay score prediction and generalization, crucial aspects of writing evaluation, such as providing feedback, have often been overlooked. Formative assessments, which aim to improve writing proficiency through ongoing interaction, necessitate providing feedback to essay authors (Havnes et al. 2012; Owen 2016). Hence, building a deep learning-based AES that not only assesses essays but also generate insightful feedback holds immense value for formative writing assessments. There are few studies that address the issue of feedback generation in AES models by incorporating traits (Hussein et al. 2020), providing explainability (Kumar and Boulanger 2020), and identifying useful sentences from essays to be used as feedback (Dong et al. 2017; Woods et al. 2017; Fiacco et al. 2022, 2023).

In this survey paper, we start by examining related surveys on deep learning-based AES. We have compiled a summary of several survey papers on deep learning-based AES published in the last five years, which is presented in Table 1. The table outlines the main themes of each survey paper, along with topics relevant to AES research such as model types, feedback generation, and dataset.

Galhardi and Brancher (2018) conducted a systematic literature review on automated short answer grading (ASAG) using classical machine learning approaches. The survey encompasses various topics such as feature selection, natural language processing and the application of machine learning in ASAG. In a related study, Haller et al. (2022) conducted a complementary survey to classical machine learning approaches, focusing on deep learning-based models and representation learning techniques. The survey delves into papers that introduce ASAG utilizing word embeddings, sequential and attention-based models. It is important to note that these surveys primarily focus on ASAG research, which differs from essay scoring as it involves the utilization of a reference answer for score calculation.

Hussein et al. (2019) and Ke and Ng (2019) discussed AES models that leverage hand-crafted features and neural network. Lagakis and Demetriadis (2021) and Ramesh and

Table 1 Comparison of survey papers on deep learning-based AES

References	Aim	ML	DL	FG	FD	EM	Datasets
Ours (2024)	Deep learning-based AES and feedback generation	✓	✓	✓	✓	✓	ASAP-AES; ASAP++; AI Hub: essay writing evaluation data; TOEFL11; ICLE
Galhardi and Brancher (2018)	Machine learning-based ASAG	✓					The University of North Texas (Texas2011); CREE and CREG from the CoMic Project; ASAP-SAS; Beetle and SciEntsBank
Hussein et al. (2019)	machine learning and deep learning-based AES	✓	✓				
Ke and Ng (2019)	Machine learning and deep learning-based AES	✓	✓			✓	
Klebanov and Madnani (2020)	Evolution of automated writing evaluation			✓	✓		
Lagakis and Demetriadis (2021)	Machine learning and deep learning-based AES	✓	✓			✓	CLC-FCE; ASAP-AES; TOEFL11; ICLE; AAE
Chassab et al. (2021)	Feature analysis; machine learning and deep learning-based AES	✓	✓				ASAP-AES; ETS corpus for non-native written English
Zhang (2021)	Automated writing evaluation systems						
Uto (2021)	Deep learning-based AES		✓				
Haller et al. (2022)	Machine learning and deep learning-based ASAG	✓	✓		✓		The University of North Texas (Texas2011); SciEntsBank; ASAP-SAS
Ramesh and Sanampudi (2022)	Machine learning and deep learning-based AES and ASAG	✓	✓		✓	✓	CLC-FCE; CREE; CS; SRA; SciEntsBank (SemEval-2013); ASAP-AES; ASAP-SAS; ASAP++; power grading; TOEFL11; ICLE
Wang et al. (2022a)	Argumentative essay scoring and feedback	✓	✓	✓	✓	✓	Stab and Gurevych (2014, 2016, 2017), Carlile et al. (2018) and Persing and Ng (2015)
Jong et al. (2023) (unpublished)	feedback providing AES systems	✓	✓	✓			ASAP; ASAP++; ICLE; AAE; RTA

The surveys are distinguished by the topics they explore, including classical Machine Learning (ML), deep learning (DL), feedback generation (FG), recommendations for future research directions (FD), evaluation metrics (EM), and datasets. A check mark (✓) indicates the inclusion of the respective topic on the survey paper

Sanampudi (2022) conducted a survey that extends the aforementioned survey papers to include transformer-based AES models. These works highlights the importance of simultaneously utilizing handcrafted features and automatically extracted features to enhance performance and generate immediate feedback. Chassab et al. (2021) discussed feature representation methods used in AES tasks. They presented a taxonomy of AES feature analysis methods, which are categorized into structure, frequency and term occurrence, semantic, and morphology feature analysis.

The study of deep learning AES models that provide scores and associated feedback is limited. As a result, few of the existing survey papers only cover feedback in relation to

deep learning-based AES models. However, Wang et al. (2022a) conducted a systematic literature review on papers that specifically focus on evaluating argumentative essays and providing associated feedback. Still this survey is limited to argumentative based feedback generation. Jong et al. (2023) conducted a review on feedback in AES where they cover few number of papers.

In this study, we conducted a survey of existing literature in deep learning-based AES models. Additionally, we extended our investigation to include automated feedback generation, which is useful for explaining the rationale behind score assignments. The organization of this article is as follows: first, we will discuss essay representation mechanisms, such as handcrafted feature engineering, one-hot encoding, and word embeddings. Next, we will cover popular deep learning methods in AES and discuss their fundamental architecture. Then, we will provide a thorough discussion on AES tasks, specifically exploring recent research categorized as holistic scoring and trait-specific scoring for prompt-specific and cross-domain prompts, as well as automated feedback generation. This work also includes widely used datasets and performance measurement metrics in AES studies. Finally, we will present the challenges and research directions in AES tasks.

2 Essay representation

AES task can be broadly divided into two phases: essay representation and essay scoring. The representation phase entails extracting features that describe the quality of the writing. To achieve this, AES utilizes methods from NLP and machine learning to extract these features. The second phase extends from the chosen essay quality features and transforms the representation into a numerical score, assumed to reflect the quality level of the essays.

In this section, we will delve into the essay representation methods utilized in the papers included in this survey.

2.1 Handcrafted features

Handcrafted feature engineering refers to the manual process of extracting features that are associated with characteristics of words, phrases, sentences, paragraphs and the entire essay. These features can vary from simple ones, such as word or essay length, to complex features like readability and coherence. Uto et al. (2020) presented several handcrafted features, including length-based, syntactic, word-based, readability, semantic, argumentation and prompt-relevant features. These features represent several feature examples, for instance, length-based features represent number of characters, words, sentences, and punctuation. Hoblos (2020) employed a semantic feature representation approach to enhance the essay representation by incorporating word meaning and sense. Besides using content-based essay representation, syntactic feature representation mechanisms like part-of-speech (POS) embedding has been utilized to build generalizable AES models (Ridley et al. 2020, 2021; Jin et al. 2018).

Handcrafted features are utilized in conjunction with classical machine learning models for score prediction, and they can also serve as a complement to deep learning models (Uto et al. 2020; He et al. 2022). Representing essays using handcrafted features comes with its

own set of advantages and disadvantages. Handcrafted features employ interpretable models for predicting essay scores, making it easier to provide feedback to students compared to using black-box deep learning models. Furthermore, syntactic and lexical features can be employed for cross domain scoring tasks, as they offer domain independent context to the model. However, manually designing features can be costly, and these features are less portable.

2.2 One-hot encoding

One-hot encoding is a simple vectorization approach in text processing. In this approach, an essay with N words is represented as sparse binary vectors. Each word in the essay is represented by a binary vector of length N , where only the index corresponding to the word is assigned a value of 1, while the rest of the elements in the vector are set to 0. One-hot representation were utilized in Taghipour and Ng (2016) to construct a lookup table layer. This representation is sparse, with all entries being zero except for the word that the vector represents.

2.3 Word embeddings

Word embedding is a dense numerical representation of words that creates distributed representations where words with similar meanings or contexts are closer together in the vector space. This representation mechanism captures the semantic meaning of words. Several recent AES research studies employ word embeddings to capture context in the essay scoring process (Cozma et al. 2018; Dong and Zhang 2016; Dong et al. 2017; Tay et al. 2018; Liang et al. 2018; Hussein et al. 2020; Song et al. 2020a).

2.3.1 C &W embeddings

C &W embeddings are trained using CNN with max-pooling. The objective of this embedding technique is to learn local context and task-specific representations by simultaneously training on multiple NLP tasks, including POS tagging and NER. Alikaniotis et al. (2016) utilize C &W embeddings to represent essays, aiming to capture semantic information before employing LSTM for essay scoring.

2.3.2 Pre-trained word embeddings

In addition to training word embeddings on the target dataset, pre-trained word embeddings can also be utilized. These pre-trained word embeddings are trained on large datasets, such as Wikipedia and news data, and can be fine-tuned on the target dataset.

2.3.3 Word2vec

Word2Vec is a software package that includes two algorithms: Continuous Bag-of-Words (CBOW) and Skip-gram (Mikolov et al. 2013). CBOW predicts a center word based on the surrounding context in terms of word vectors, while the skip-gram algorithm predicts the

distribution of context words based on a center word. These algorithms learn word embeddings by considering local context windows.

2.3.4 GloVe

GloVe utilizes both global and local context windows to generate word embeddings. This word embedding technique is trained on the nonzero elements of a word-word co-occurrence matrix (Pennington et al. 2014). Uto et al. (2020) employed GloVe to capture context information from essays, which was used in conjunction with handcrafted features for essay scoring. Cao et al. (2020) utilized 200-dimensional GloVe vectors to initialize the word embeddings in the HA-LSTM architecture for encoding essays. Jin et al. (2018) utilized GloVe to capture semantic information from source prompts, filtering out extreme essays to train a prompt-dependent model.

2.3.5 FastText

FastText is an extension of Word2Vec that provides embeddings at both the character and word-level (Bojanowski et al. 2016). This type of embedding can be beneficial for generating embeddings for out-of-vocabulary (OOV) words.

2.3.6 POS embedding

Ridley et al. (2020, 2021) proposed the utilization of part-of-speech (POS) embeddings to learn syntactic representations of essays and apply them for cross-prompt essay scoring.

3 Deep learning

The use of deep learning in AES eliminates the need for traditional feature engineering, enabling AES models to score essays in an end-to-end fashion. In this section, we will provide the highlight of widely used deep learning models for the AES task.

3.1 Convolutional networks

A convolutional neural network (CNN) (LeCun et al. 2015) is a specific type of feedforward neural network architecture designed for processing data with grid-like topology, such as images. This architecture comprises three main types of layers: convolutional layer, pooling layer, and fully-connected (FC) layer. While each layer type serves a different purpose, we will specifically focus on the utilization of the convolutional layer in AES tasks. The convolutional layer serves as the initial layer in the network and is responsible for detecting patterns in essays. Several recent studies have proposed the use of the convolutional layer to capture local contexts from large sequences of essays (Taghipour and Ng 2016; Hussein et al. 2020; Dong and Zhang 2016; Dong et al. 2017; Liang et al. 2018; Ridley et al. 2020, 2021). A convolution for segment (window), $\bar{x}$ in an essay, $Conv(\bar{x})$, is computed as follows:

$$Conv(\bar{x}) = W \cdot \bar{x} + b,$$

where W is the filter, b the bias and $\bar{x}$ is an input window. The window slides across the essay, collecting all the feature maps that can ultimately be utilized for score prediction.

3.2 Recurrent networks

Recurrent neural networks (RNNs) are a type of neural network specifically designed for processing sequential data. Their architecture enables the propagation of information across a sequence by maintaining an internal hidden state. This allows RNNs to capture dependencies and patterns from previous states, influencing the decisions made at subsequent states.

$$\begin{aligned} H_t &= W_{xh}X_t + W_{hh}H_{t-1} + b_h \\ O_t &= W_{ho}H_t + b_o \end{aligned}$$

where $X_t \in \mathbb{R}^{n \times d}$ is an input at time step t , $H_t \in \mathbb{R}^{n \times h}$ is the hidden layer output of time step t , H_{t-1} is the hidden layer output from previous time step and O_t is the output at time step t . $W_{xh} \in \mathbb{R}^{d \times h}$, $W_{hh} \in \mathbb{R}^{h \times h}$, and $W_{ho} \in \mathbb{R}^{h \times o}$ are the learnable weights of the input, hidden state and output layer respectively. $b_h \in \mathbb{R}^{1 \times h}$ and $b_o \in \mathbb{R}^{1 \times o}$ are biases of the hidden and the output layers, respectively. RNNs can suffer from the problems of vanishing and exploding gradients. To address these issues, a variant of RNN called Long Short-Term Memory (LSTM) was introduced. LSTM (Hochreiter and Schmidhuber 1997) incorporates a memory cell, which enables the network to learn when to forget, ignore, or store information in the memory cell. The forget gate (f_t) allows the memory cell to reset the cell state, the input gate (i_t) is responsible for how much of the input node's ($\tilde{c}_t$) value should be added to the cell state (c_t). The output gate (o_t) determines how to update the values of the hidden units. This mechanism helps mitigate the vanishing and exploding gradient problems in RNNs. The functions of LSTM are computed as follows

$$\begin{aligned} i_t &= \sigma(W^{(i)}x_t + U^{(i)}h_{t-1} + b_i) \\ f_t &= \sigma(W^{(f)}x_t + U^{(f)}h_{t-1} + b_f) \\ o_t &= \sigma(W^{(o)}x_t + U^{(o)}h_{t-1} + b_o) \\ \tilde{c}_t &= \tanh(W^{(c)}x_t + U^{(c)}h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\ h_t &= o_t \odot \tanh(c_t) \end{aligned}$$

x_t and h_t are the input and output at step t , respectively. $W^{(i)}$, $W^{(f)}$, $W^{(o)}$, $W^{(c)}$, $U^{(i)}$, $U^{(f)}$, $U^{(o)}$, and $U^{(c)}$ are weight matrices and b_i , b_f , b_o , and b_c are bias vectors. The $\odot$ and σ symbols represent element-wise multiplication and the sigmoid function, respectively. LSTM is a widely used neural network approach in AES task research. Its capability to capture dependency information from long sequential inputs has made it popular in various NLP tasks. In the AES task, LSTMs are utilized to capture global context information from essays (Taghipour and Ng 2016; Tay et al. 2018; Ridley et al. 2020, 2021; Alikaniotis et al. 2016; Dong and Zhang 2016; Liang et al. 2018; Uto et al. 2020, 2020; Hussein et al. 2020).

3.3 Attention mechanism

The attention mechanism (Bahdanau et al. 2014, 2014) is an architecture that assigns a weight to each hidden state representation of an input sequence in order to determine which part of the input to focus on. One advantage of the attention mechanism is its ability to parallelize computations, as it does not rely on the sequence order. This mechanism has become a core component of transformer models.

$$h_i = \sum \alpha_i v_i$$

$$\alpha_i = \frac{e^{(W_u \cdot m_i)}}{\sum e^{(W_u \cdot m_j)}}$$

where α_i and v_i are attention weights and representation of the input (hidden states from lower layers), respectively. W_u represents the weight vectors and m_i represents the attention vectors for the i -th word, and h_i is the final representation of weighted sum of all the word vectors. Integrating attention mechanisms with CNN and LSTM has emerged as a popular approach in various research studies (Taghipour and Ng 2016; Dong et al. 2017; He et al. 2022). This approach enables the models to capture the importance of different words or elements to attend to, resulting in significant performance improvements and providing interpretable results.

3.4 Transformer-based models

CNNs and RNNs primarily focus on a limited scope of word-level information, making it challenging to capture long-distance dependencies and learn deep semantics. In contrast, recent studies proposed the utilization of pre-trained models, such as BERT, to represent the contextual information of essays with rich semantic understanding. Pre-trained models are deep learning models trained on extensive datasets and can be used as-is or fine-tuned for specific tasks.

BERT (Devlin et al. 2018), which stands for Bidirectional Encoder Representations from Transformers, is a pre-trained bidirectional encoder-only transformer. It is trained on unlabeled text, considering both left and right context in all layers. BERT undergoes pre-training on two target tasks: masked language modeling and next sentence prediction. In the context of AES tasks, BERT is used as an initial essay representation module, which is then fine-tuned on the essay training set.

By introducing an additional layer to BERT, the model parameters can be updated to accommodate the essay scoring training data. Several research studies utilize fully-connected layer on top of the BERT-encoded essay to generate a score (Yang et al. (2020); Xie et al. (2022); Wang et al. (2022b); Xue et al. (2021)).

4 Automated essay scoring

In this section, we introduce prompt-specific and cross-prompt scoring subsections. Within each subsection, we present papers on trait-specific and holistic scoring.

4.1 Prompt-specific automated essay scoring

Prompt-specific AES models are models that are trained and tested within the same domain. These models learn to capture the specific features of an essay topic based on a given prompt, enabling them to provide evaluations that are more specific to the domain. Such models are particularly useful in assessing essay quality traits such as prompt adherence and coherence.

4.1.1 Holistic scoring

Prompt-specific holistic scoring AES models refer to AES models that are trained and tested within the same domain and predict an overall score. In this section, we explore various studies that focus on holistic scoring approaches.

4.1.2 Neural network-based models

The earliest application of deep learning models in AES tasks can be attributed to Taghipour and Ng (2016). This study introduced a neural network approach that predicts essay scores without the need for feature engineering. As it is shown in Fig. 1, the architecture utilizes CNN and LSTM to capture both local and global context information. Linear regression and sigmoid activation functions were used for the holistic score prediction. However, this work has certain limitations in terms of essay representation. It relies on one-hot encoding, which is insufficient for capturing rich semantic information and leads to sparse representation. Furthermore, it treats the essay as a sequence of words, overlooking the hierarchical word, sentence, and paragraph levels in the actual document structure of the essay. Several research works have proposed solutions to address these limitations, while others have extended this work to enhance scoring accuracy in AES models.

In Table 2, we present a summary of neural network-based prompt-specific holistic scoring papers, with many of them being extensions of this work.

Introducing semantics using word embeddings: Alikaniotis et al. (2016) proposed using C & W word embeddings for essay representation instead of one-hot encoding to capture score-specific word senses and dependencies. The inclusion of word embeddings allows the

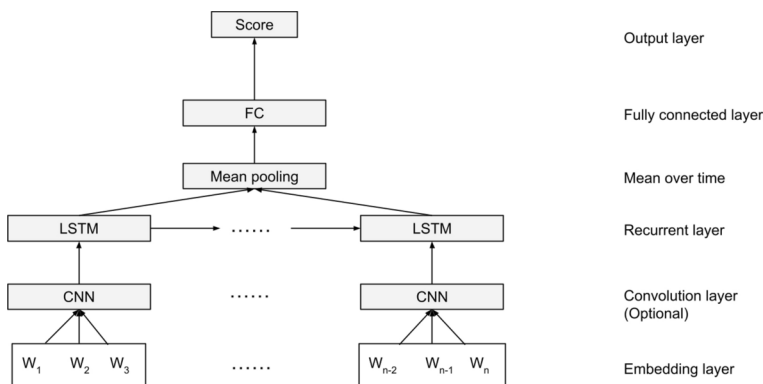


Fig. 1 Convolutional recurrent neural network AES architecture (Taghipour and Ng 2016)

Table 2 A summary of neural network-based prompt-specific holistic AES studies

References	Main theme	Essay representation					Datasets
		SW	DM	CM	HF	RF	
Taghipour and Ng (2016)	An LSTM with mean-over-time	✓					ASAP-AES
Alikaniotis et al. (2016)	An LSTM model with score-specific C & W word embeddings	✓					ASAP-AES
Dong and Zhang (2016)	A hierarchical CNN-CNN structure		✓				ASAP-AES
Dong et al. (2017)	A hierarchical CNN-LSTM structure with attention pooling		✓				ASAP-AES
Dasgupta et al. (2018)	A CNN-LSTM architecture augmented with textual quality features		✓		✓		ASAP-AES
Liang et al. (2018)	A Siamese CNN-LSTM architecture that captures information from essay and ratings		✓			✓	ASAP-AES
Wang et al. (2018)	A dilated LSTM model that augments rating schema using reinforcement learning	✓				✓	ASAP-AES
Uto and Okano (2020)	A hierarchical neural network with IRT that handles bias from ratings		✓			✓	ASAP-AES
Farag et al. (2018)	A CNN-LSTM with local coherence model		✓	✓			ASAP-AES
Mesgar and Strube (2018)	An LSTM with coherence modeling	✓		✓			ASAP-AES
Tay et al. (2018)	A CNN-LSTM with neural coherence features		✓	✓			ASAP-AES
Li et al. (2018)	An attention-based coherence scoring model with LSTM	✓		✓			ASAP-AES, CEEE

The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure (DM), using coherence modeling (CM), incorporating handcrafted features (HF) and incorporating rating features (RF)

model to consider semantic information within the essay. However, this work still treated the essay text as a sequence of words.

Document modeling using hierarchical architecture: Dong and Zhang (2016) proposed a two-layered CNN hierarchical architecture (Fig. 2) to capture word-level and sentence-level contexts from essays, aiming to model a document structure. In this architecture, the input essay is segmented into sentences ($S^1, \dots, S^n$), with the first CNN layer capturing word-level contexts and the second CNN layer capturing sentence-level contexts.

Expanding on this work, Dong et al. (2017) proposed an AES architecture that utilizes CNN, LSTM and attention mechanisms to model document structure. They employed CNN for sentence representation and LSTM for document representation, enabling the capture of contexts from both individual sentences and the entire document. Furthermore, attention pooling was utilized after the convolutional layer and LSTM layer to identify the relevant words and sentences within the essay.

Augmenting text quality features: Instead of solely focusing on word and sentence representation in essays, several studies have shown the impact of incorporating additional features to enhance the performance of AES model (Dasgupta et al. 2018; Liang et al. 2018). Dasgupta et al. (2018) proposed qualitatively enhanced convolutional recurrent neural net-

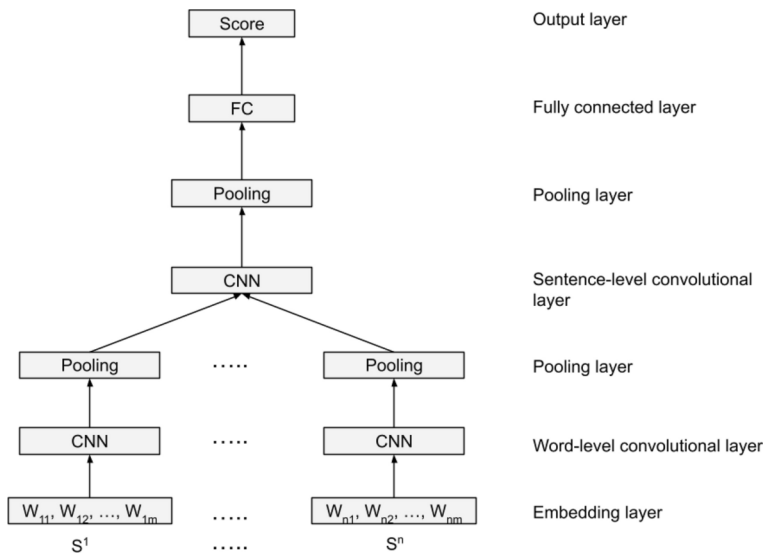


Fig. 2 Hierarchical CNN AES architecture (Dong and Zhang 2016)

works that learn from both latent linguistic characteristics of the essay and automatically identified complex patterns using neural networks. This method enhances the convolutional recurrent neural network by recognizing the interconnections among the various factors involved in assessing the quality of text. The linguistic features used to create linguistically informed models include part of speech (POS), universal dependency relations, structural well-formedness, lexical diversity, sentence cohesion, causality, and informativeness of the text.

Incorporating rating information: The consideration of rating information in the design of AES methods plays a significant role in enhancing the performance of the neural models. However, representing rating information and fuse it with essay content has been a challenge in the process of designing AES models.

Liang et al. (2018) proposed a Siamese bidirectional LSTM architecture to capture semantic features of the essays and the rating criteria information embedded within them. As it is shown in Fig. 3, the input essay is represented using a convolutional recurrent neural network, which produces context vectors and sentence similarity vectors. The context vectors represent the semantic features of the essay, while the sentence similarity vectors reflect the coherence within the essay.

Furthermore, this study incorporates rating criteria information by introducing sample essays. The rating criteria within the essays (cross-feature) are captured through contrastive learning between the input essay and the samples. The cross-feature details are conveyed through distance information extracted from the input and sample essays. In essence, the final essay score in the Siamese Bi-LSTM architecture takes into account the essay's semantic features, coherence, and the underlying rating criteria information within the essays.

In another note, Wang et al. (2018) proposed a reinforcement learning approach that utilizes quadratic weighted kappa as guidance to optimize the essay scoring system, capturing rating information.

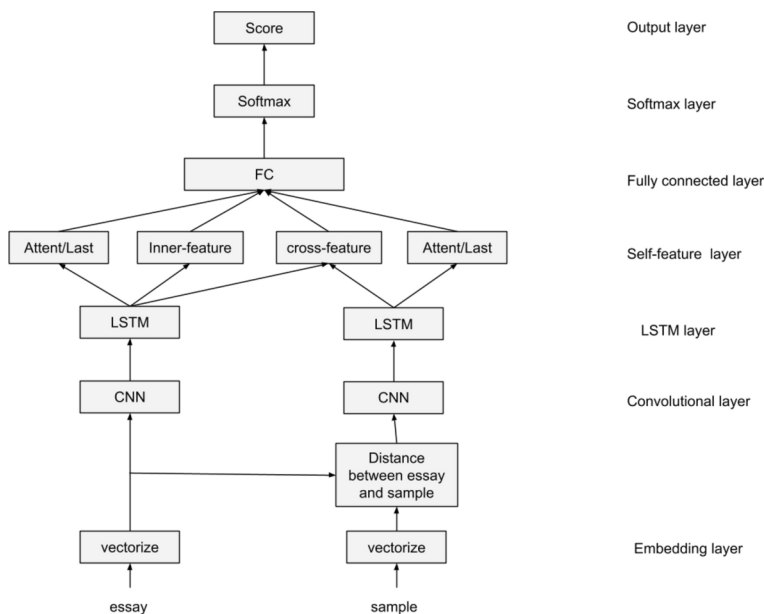


Fig. 3 Siamese Bi-LSTM AES architecture (Liang et al. 2018)

Uto and Okano (2020) investigated the performance of AES methods in accordance with the characteristics of raters inscribed implicitly in the ratings. To build a robust neural network AES methods, they proposed a two-stage architecture that uses Item Response Theory (IRT) model alongside the convolutional recurrent neural networks. The IRT model is used to model the relationship between the raters and the essays and handle the possible bias that can be introduced in the scores. Besides building robust models, IRT models demonstrated the capability to enhance the performance of AES models in both convolutional recurrent neural networks and BERT (Uto et al. 2023).

Incorporating neural coherence features: Farag et al. (2018) highlighted the limitation of convolutional recurrent neural network AES models when faced with adversarially crafted grammatically correct inputs consisting of incoherent sequences of sentences. Several AES studies have suggested integrating coherence features to enhance model performance and detect such adversarially crafted texts.

Mesgar and Strube (2018) proposed a local coherence model that identifies salient information and extracts patterns of changes as a text progresses. The model captures the representation of salient information by measuring the similarity between LSTM hidden states of adjacent sentences. Semantic shifts are captured by applying CNN on the similarity measure of salient information between sentences. This study underscores the significance of coherence features by augmenting the local coherence model features with EASE (Enhanced AI Scoring Engine),¹ which relies on manually crafted features.

Furthermore, Farag et al. (2018) proposed a local coherence model that measures sentence fluency by building LSTMs over sentences in an essay and applying a convolutional layer over the last output of the LSTM layer. Through a linear operation followed by a

¹ EASE is an open source AES system.

sigmoid activation function, a coherence score is predicted. By simultaneously training on local coherence and overall score, the model predicts AES scores and identifies adversarial inputs.

Tay et al. (2018) proposed the SKIPFLOW mechanism to capture both the semantic features of an essay and the semantic relationship between two points within an essay (coherence). They utilized LSTM with average pooling to represent the essay and neural tensors to model the semantic relationship between two snapshots of hidden representations from the LSTM layer. As it is shown in Fig. 4, the neural coherence features and the global context information are fed into a linear layer followed by sigmoid activation function. The SKIPFLOW mechanism captures global context information and models coherence, however, it overlooks the importance of document modeling.

Li et al. (2018) proposed a coherence-informed essay representation using multi-head self-attention mechanism. An essay is represented using word-to-word relationships by multi-head self-attention mechanism. After obtaining this relationship-based essay representation, LSTM with average pooling is employed to generate the overall score. This architecture differs from the aforementioned coherence modeling approaches in that coherence is embedded directly into the essay representation, rather than treating coherence information as supplementary input to the scoring model.

4.1.3 Transformer-based models

Employing BERT in NLP tasks has demonstrated the extraction of high-quality contextualized word embeddings rich in semantic information. This high-quality contextual information enhances model performance and makes transformer-based models the preferred approach in NLP research. Table 3 presents a summary of transformer-based AES studies with the core theme, essay representation and employed datasets.

A simple AES BERT architecture, as depicted in Fig. 5, takes an essay text as an input and feeds it to the Transformer Encoder. Subsequently, BERT generates word embeddings for each token and a fixed-size representation of the entire script, denoted by $[CLS]$. The $[CLS]$ token encapsulates the comprehensive representation of the document as contextual word embeddings. By employing a fully connected linear layer (FC) with a sigmoid activation function, a score for a given essay can be predicted.

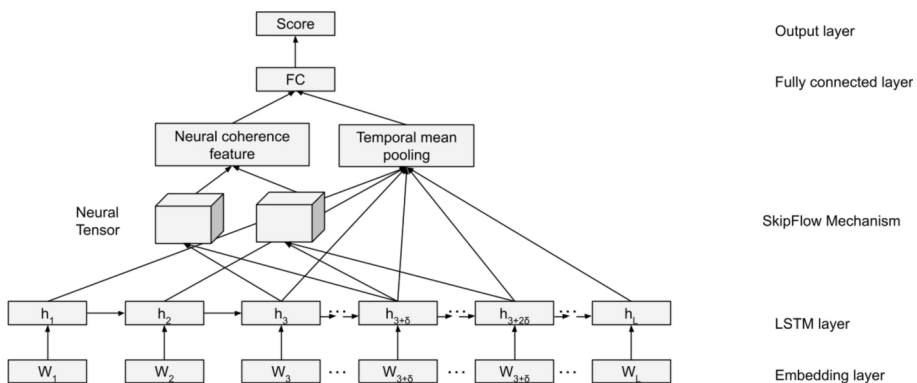


Fig. 4 SKIPFLOW LSTM with width σ AES architecture (Tay et al. 2018)

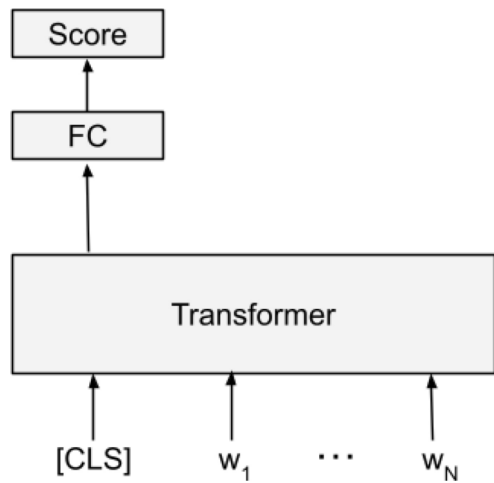
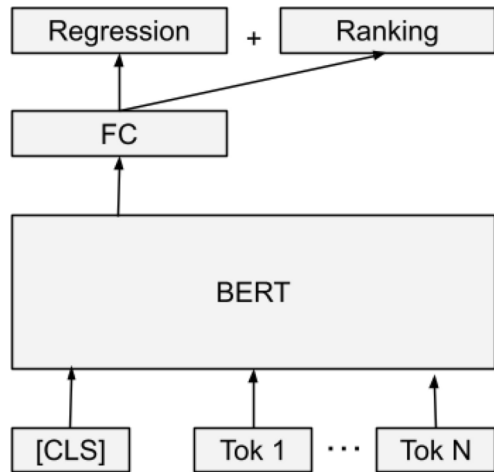
Table 3 A summary of transformer-based prompt-specific holistic AES studies

References	Main theme	Essay representation			Datasets
		SW	DM	HF	
Rodriguez et al. (2019)	Fine-tune BERT	✓			ASAP-AES
Nadeem et al. (2019)	Fine-tune BERT	✓			ASAP-AES
Mayfield and Black (2020)	Fine-tune BERT	✓			ASAP-AES
Yang et al. (2020)	Fine-tune BERT with multi-loss methods for optimizing the learning function	✓			ASAP-AES
Xie et al. (2022)	A neural pairwise contrastive leaning with BERT for essay representation	✓			ASAP-AES
Wang et al. (2022b)	Multi-scale essay representation with BERT		✓		ASAP-AES
Uto et al. (2020)	Fine-tune BERT with essay-level features	✓		✓	ASAP-AES
Li et al. (2023)	Fine-tune BERT with sentence and essay-level features		✓	✓	ASAP-AES
Cho et al. (2023)	Group essays based on their rubrics and represent essay using BERT	✓			ASAP-AES
Mizumoto and Eguchi (2023)	Prompting GPT-3 for essay scoring				TOEFL11
Yancey et al. (2023)	Prompting GPT 3.5 and GPT-4 for essay scoring and feedback generation				ASAP-AES

The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure(DM), using coherence modeling (CM), and incorporating handcrafted features (HF)

Mayfield and Black (2020) investigated whether fine-tuning BERT leads to better performance compared to convolutional recurrent neural networks in AES tasks. The study found that BERT-based models achieved results comparable to those of neural network models. However, the parallelization and pre-trained knowledge did not offer significant advantages over the CNN and RNN based models. Moreover, several other studies have reported similar results with BERT-based models (Rodriguez et al. 2019; Nadeem et al. 2019).

On the contrary, Yang et al. (2020) demonstrated the utilization of BERT for improved performance results by extending its capabilities to predict both scores and ranks. Fine-tuning of BERT with multi-loss functions (Yang et al. 2020; Xie et al. 2022; Wang et al. 2022b) and augmenting BERT with essay-level features (Uto et al. 2020) have yielded results that surpass the state-of-the-art neural network methods in AES tasks.

Fig. 5 Simple transformer-based AES architecture**Fig. 6** R²BERT framework for AES (Yang et al. 2020)

Fine-tuning BERT with multi-loss functions: Yang et al. (2020) proposed R²BERT, a framework that leverages BERT for essay representation and incorporates multi-loss methods, including regression loss and ranking loss, to optimize the learning process. This study viewed AES task as both a regression and ranking problem. The rationale behind this approach is that assigning grades and rankings are complementary tasks in real-world exam evaluation scenarios.

As it is shown in Fig. 6, R²BERT employs BERT for essay representation, and then applies fully connected linear layer followed by sigmoid function on the fixed essay representation, [CLS]. After predicting the score, both the accuracy and rank information are utilized to optimize the learning process. To achieve this, R²BERT combines both mean squared error and cross-entropy loss functions. The mean squared error, serving as the regression loss, ensures the accuracy of the score in each batch, while the cross-entropy loss, acting as the ranking loss, preserves the order within the batch. However, the order

information captured from the batch overlooks the essay's rank within the entire dataset, impacting the model's accuracy.

Although R₂BERT demonstrated superior performance compared to convolutional and recurrent neural networks, it has several limitations: (1) batch-wise ranking overlooks global order information, (2) determining the appropriate weights to balance the loss functions for optimization, and (3) BERT only supports sequences of up to 512 tokens, leading to the truncation of longer essays.

Xie et al. (2022) proposed a neural pairwise contrastive learning (NPCR) approach that addresses the issues with global ranking and weighting the objective functions. NPCR addresses these issues by introducing neural pairwise ranking instead of batch-wise ranking in order to capture rank information from the entire dataset. As it is shown in Fig. 7, the NPCR approach learns essay representation by comparing the input essay with sample essays. This is achieved by employing neural pairwise ranking model that learns both order information and relative semantic representation. The semantic representation is the difference between BERT representation of the input and reference essays, defined as relative scores. The predicted score is then obtained by averaging the sum of the relative scores and the reference score. NPCR uses a multi-sample voting strategy to determine the final score. The use of neural pairwise ranking addresses the two issues that arise from the use of batch-wise ranking. However, the limitation of capturing contexts from long sequence essay inputs due to BERT's 512 token size is not addressed.

Wang et al. (2022b) proposed a joint learning of multi-scale essay representation mechanism for AES tasks with BERT. This study highlights the importance of capturing token-level, segment-level and document-level information and optimizing the learning function using regression and ranking functions. As it is shown in Fig. 8, multi-level attributes of essays are captured by extracting token-level and document-level multi-scale features from the essay representation of BERT. The contextualized word embeddings of each token is used as token-level representation, and the fixed representation is used as the entire essay representation. These two types of information are concatenated and fed into a dense regression model to predict a score corresponding to the token- and document-level of the essay representation.

To overcome the token-length constraint in BERT, this study introduces segment-level essay representation. This is accomplished by dividing essays into segments of equal token length and applying independent BERT models to each segment. By utilizing LSTM and subsequent attention pooling on the fixed representations of these segments, a sequential contextual information is modeled, which is utilized for score prediction after employing a fully connected neural network. The final essay score is determined by combining the scores derived from token-level and document-level representations with the score from segment-level representation. While the segment-level representation of essays prevents the truncation of essay sections, representing essay segments solely as token sequences lacks the concept of document structure.

Augmenting BERT with essay-level features: Uto et al. (2020) proposed a hybrid approach that integrates essay-level features with BERT. This work incorporates various essay-level features, including length-based, syntactic, word-based, readability, semantic, argumentation, and prompt relevance features. These features are concatenated with BERT representation of the essay and passed through a fully connected linear layer as it is shown in Fig. 9.

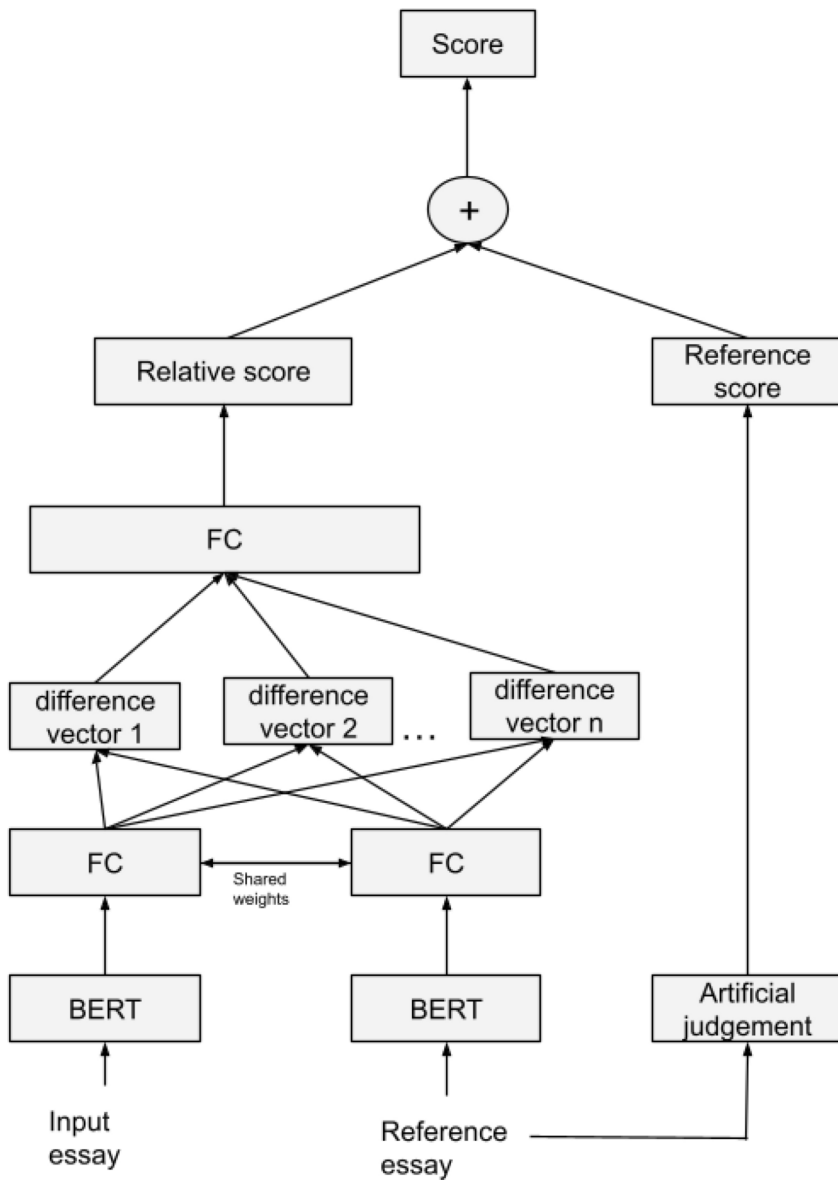


Fig. 7 The overall framework of neural pairwise contrastive regression (NPCR) model for AES (Xie et al. 2022)

Similarly, Li et al. (2023) proposed a framework that incorporates multi-scale features, including handcrafted features, sentence-level features, document-level features and prompt-relevance features. As it is shown in Fig. 10, document-level features are extracted using LSTM with average pooling as proposed by Taghipour and Ng (2016), while the sentence-level features are captured by employing sentence-BERT (SBERT) on the essay, followed by LSTM with attention pooling.

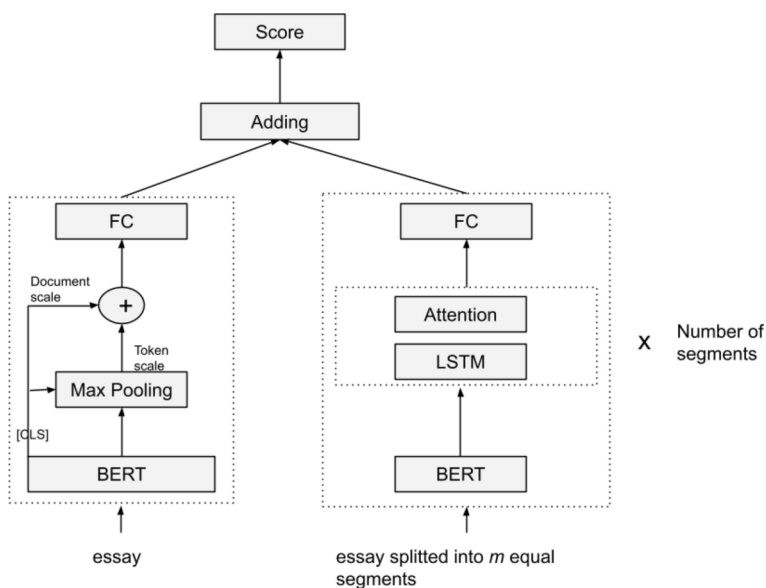
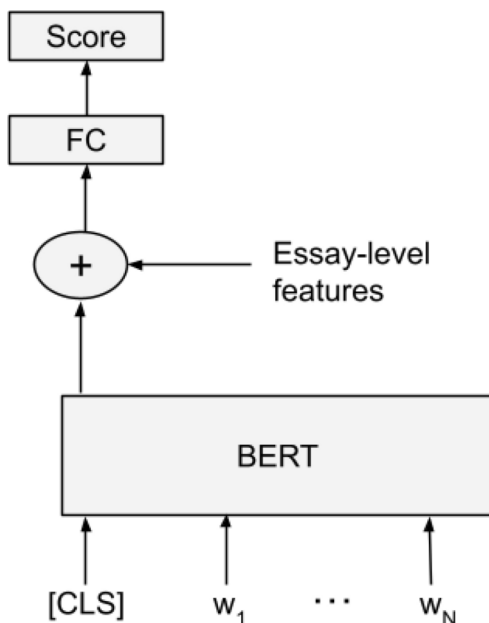


Fig. 8 Multi-scale essay representation AES architecture (Wang et al. 2022b)

Fig. 9 BERT-based model with essay-level features AES architecture (Uto et al. 2020)



Prompt adherence is computed by comparing the prompt and essay representations for similarity. The method employs Doc2Vec to represent prompts and essays, using cosine similarity to measure the degree of relevance. The handcrafted features include aspects like grammatical correctness, text richness and other length-based features. The final essay score

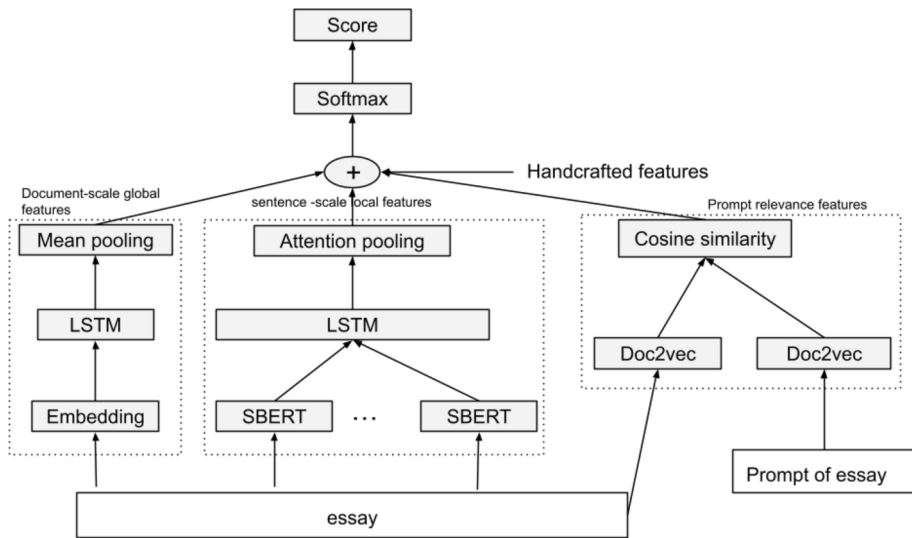


Fig. 10 Multi-scale features design for AES (Li et al. 2023)

is derived by combining the sentence-level, document-level, prompt adherence, and handcrafted features, utilizing a fully connected linear layer with a sigmoid activation function.

While manual feature engineering has its limitations, integrating with BERT-based models demonstrates a substantial performance improvement. BERT-based models introduce rich contexts to the essay representation, ultimately enhancing the overall performance of the models.

Rubric-specific training models: Cho et al. (2023) proposed the adoption of rubric-based training models. Instead of building prompt-specific models, their work introduced a rubric-specific model. Prompt-specific models tends to overlook various attributes that are integral to the rubrics. Rubrics provide criteria to be considered during the grading of students' essay. Rubric-specific training models demonstrated the ability to capture features from essays that are shared across different prompts. Attributes such as the student's grade level serve as important indicators in scoring essays, enabling adjustments based on the writing ability within the same grade level. Such valuable information is encoded within the rubrics. However, prompt-specific models fail to account for these attributes.

GPT for holistic score: Several studies investigated the use of large language model for AES (Mizumoto and Eguchi 2023; Yancey et al. 2023; Mizumoto and Eguchi 2023, 2023) conducted a study exploring the application of GPT-3 for AES on the TOEFL11 dataset (see Sect. 6.4). The authors examined the GPT model's ability to score essays by inputting the prompt, rubric and essay. The model then predicts a score category of Low, Medium, or High. Furthermore, the study investigated the influence of linguistic features in conjunction with GPT. In a similar manner, Yancey et al. (2023) investigated the employment of GPT-3.5 and GPT-4 for AES. The integration of GPTs in AES showed a promising accuracy levels, indicating their potential for scoring essays and generating feedback. However, if the model is transformed into a regression task aimed at producing a numerical score, the performance of the model is expected to significantly deteriorate.

4.1.4 Trait-specific scoring

This section refers to models that employ the same domain for both training and testing, and assign scores to the various traits associated with essay quality. The overall essay score can be calculated by applying a weighted sum function to all the traits.

4.1.5 Neural network-based models

Hussein et al. (2020) introduced a trait-specific adaptation of the method proposed by (Taghipour and Ng 2016). They extended the convolutional recurrent neural network, depicted in Fig. 1, to predict both the overall score and the traits scores, as it is seen on Fig. 11. In this extension work, the limitations observed on the original work still exists, which are lack of semantic representations and document modeling.

In Table 4, we present a summary of neural network-based prompt-specific trait-specific AES studies with essay representation approaches and datasets.

Many hands make light work: Kumar et al. (2021) proposed a similar approach that addresses the aforementioned limitations by incorporating pre-trained word embeddings and document modeling. In this study, the authors place particular emphasis on designing prompt-specific and trait-tailored scoring from embedding to trait score. Each trait score and the overall scores are trained on the input essay but obtain different weights and parameters as it is shown in Fig. 12. Each trait score can be modeled as it fits to the trait definition. For example, argumentative or narrative essays do not possess a prompt adherence trait, which means that prompts designed for argumentative or narrative essay types will not be scored based on the prompt adherence trait quality.

Enhancing traits with trait-specific features: He et al. (2022) proposed a Hierarchical Multi-task Trait Scorer (HMTS) for the Chinese AES task. The architecture consists of two layers: the common layer and the private layer. The common layer is dedicated to learning sentence and paragraph representations. To achieve this, the XLNet encoder is employed for sentence representation, while the Bi-LSTM and attention pooling layer are utilized to encode and learn paragraph representations.

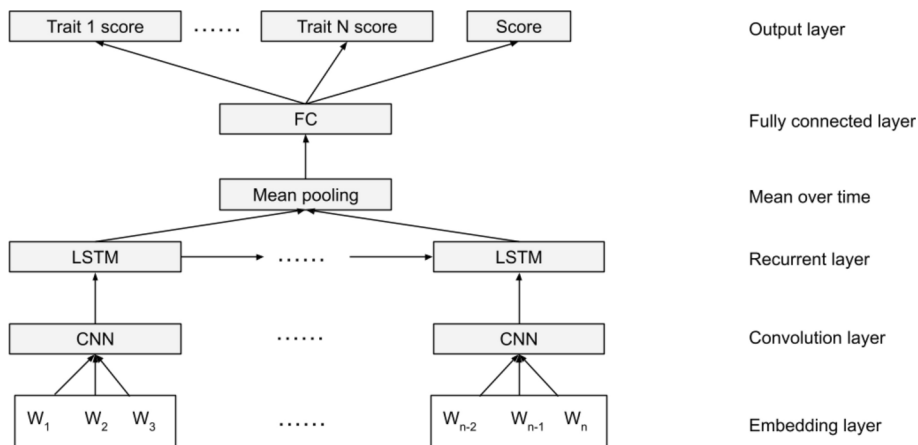
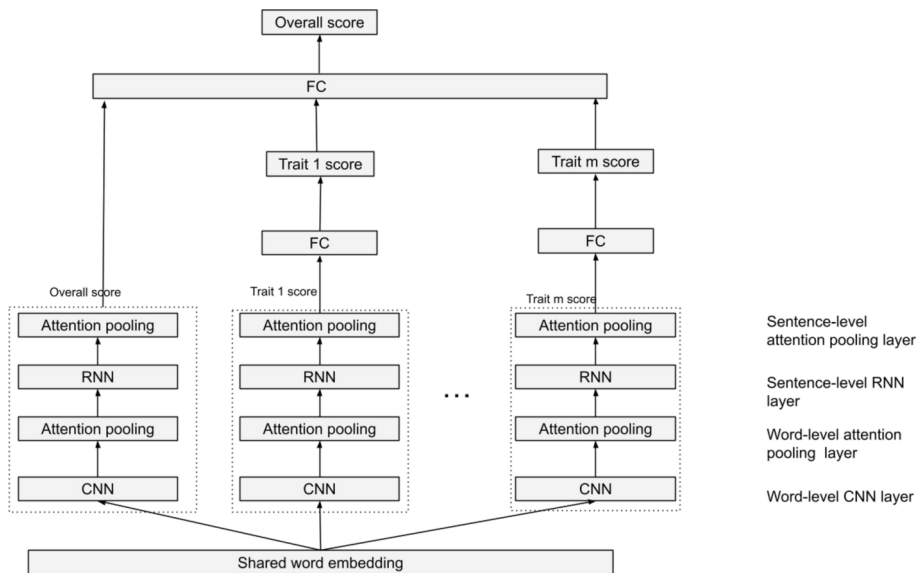


Fig. 11 LSTM-based trait-specific AES (Hussein et al. 2020)

Table 4 A summary of neural network-based prompt-specific trait scoring AES studies

References	Main theme	Essay representation			Datasets
		SW	DM	HF	
Hussein et al. (2020)	a Convolutional recurrent neural network for trait specific scoring		✓		ASAP-AES
Kumar et al. (2021)	A multi-task learning convolutional recurrent neural network for trait specific scoring		✓		ASAP-AES
He et al. (2022)	A hierarchical multi-task learning convolutional recurrent neural network with trait-specific features for trait specific scoring		✓	✓	ACEA
Song et al. (2020a)	A hierarchical multi-task learning for organization evaluation using LSTM and CNN	✓			Chinese argumentative corpus

The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure (DM), using coherence modeling (CM), and incorporating handcrafted features (HF)

**Fig. 12** Trait-specific scoring using multi-task learning (Kumar et al. 2021)

The private layer, on the other hand, is responsible for learning trait-specific representations. This layer incorporates trait-to-trait interaction and augments traits with more relevant features to enhance the essay representation for each task. The work emphasizes the significance of enriching the original essay representation with trait-specific features. The interaction among traits is captured using attention mechanisms, resulting in improved overall and trait-specific scores. However, preparing manually crafted trait-specific fea-

tures for prompts and traits requires an extensive work. Besides the drawbacks introduced from the use of LSTM and XLNet, we identified that handcrafted features pose challenges in terms of preparation and building models that can be applied to various forms of prompts and traits.

Organizational quality of essays: Evaluating the organization quality in essay writing allows us to measure the logical progression of arguments and coherence within the essay. However, modeling essay organization faces challenges due to the limited capability of models to represent the essay's organization and relate that representation to organization quality. In their work, Song et al. (2020a) proposed an approach to represent and evaluate the notion of organization in argumentative essays using sentence- and paragraph-level discourse elements. These elements help identify the functions of sentences and paragraphs, such as introduction, thesis, main idea, evidence, elaboration, or conclusion.

In this study, the authors designed a multi-task model that jointly learns sentence function identification and paragraph function identification for organization evaluation. As it is shown in Fig. 13, Bi-LSTM is used to represent sentence-level essay features, and positional information is incorporated to form a representation of sentence functions. At the paragraph-level, the paragraph identification function utilizes Bi-LSTMs to organize the contextual sentence representations into a paragraph structure. To predict both the sentence functions and paragraph functions, softmax is applied over the function representations. Finally, the organization quality of the essay is represented as a tensor representation that encompasses both sentence functions and paragraph functions. By applying CNN and max pooling over the tensor, a score for the organization trait is generated.

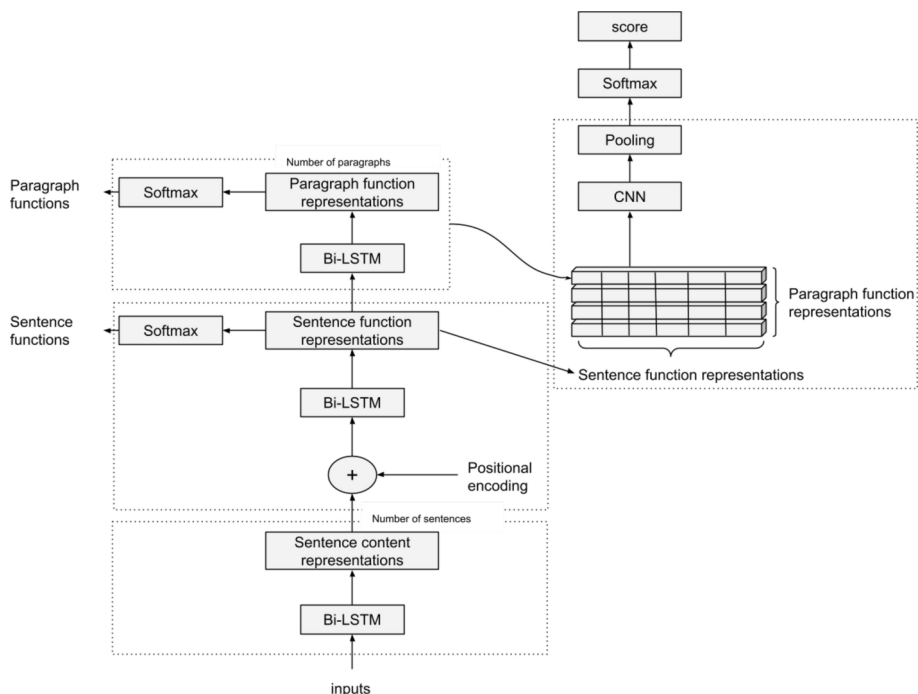
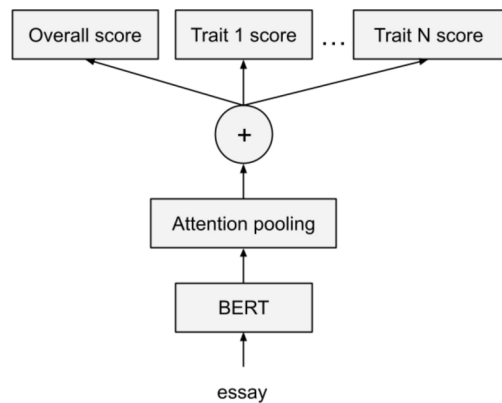


Fig. 13 Hierarchical multi-task learning for organization evaluation architecture (Song et al. 2020a)

Table 5 A summary of transformer-based prompt-specific trait AES papers

References	Main theme	Essay representation			Datasets
		SW	DM	HF	
Xue et al. (2021)	Hierarchical multi-task learning approach with essay represented using BERT		✓		ASAP-AES
Hicke et al. (2023)	Persuasive quality of argumentative essays using DeBERTa		✓		Not available

The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure(DM), using coherence modeling (CM), and incorporating handcrafted features (HF)

Fig. 14 Hierarchical BERT-based trait-specific scoring for AES (Xue et al. 2021)

Overall, this hierarchical multi-task learning approach enables the evaluation of the organizational quality of argumentative essays by optimizing sentence and paragraph functions, leading to improved assessment of the essays' organizational traits.

4.1.6 Transformer-based models

In this section, we present studies that proposed transformer-based prompt-specific trait scoring AES. Table 5 presents the summary of the studies with essay representation mechanisms and datasets employed.

Hierarchical multi-task learning for trait scoring: Xue et al. (2021) developed a hierarchical multi-task learning approach that involves fine-tuning BERT for multi-dimensional AES tasks. In order to handle lengthy essays, the method divides the input text into segments with a length of 510 tokens. As it is shown in Fig. 14, BERT is employed to represent these essay segments. By combining the representations of all the essay segments, a fully connected linear layer is utilized to predict both holistic and trait-specific scores. The study demonstrates that fine-tuning BERT for trait-specific scores using multi-task learning improves the performance of AES models.

However, there are some limitations to consider in this study. Firstly, the essay script is treated as a sequence of words, which overlooks the potential information that could be

derived from sentence-, paragraph-, or document-level structures. Secondly, the model predicts a holistic score as an independent trait, which does not align with the evaluation methods typically used in classroom settings. In practice, the holistic score is usually a weighted sum of the individual trait scores.

Persuasive quality of argumentative essays: In argumentative essays, considering the level of persuasiveness is an important aspect when determining the overall score. However, capturing the level of persuasiveness from argumentative essays is not a simple task. The use of discourse elements in argumentative essay demonstrated the capability of picking organization quality features (Song et al. 2020a). In a similar manner, Hicke et al. (2023) utilizes discourse element features to identify persuasive sections in argumentative essay. They employed DeBERTa for a robust essay representation. Each token from this representation is scored its level of effectiveness in its corresponding discourse element. The average of token scores is used as overall score of the discourse element.

The model predicts the persuasive degree of discourse elements, enabling the provision of feedback that specifically focuses on the organizational quality of these discourse elements. Models like these show promise in enhancing feedback mechanisms for essays by considering the quality and persuasive nature of the sentences.

4.2 Cross-prompt automated essay scoring

In this section of the survey, we cover studies that propose models trained and tested in different domains, which are referred to as cross-domain or cross-prompt. These articles conduct experiments on both prompt-specific and cross-prompt settings. However, since there is less focus on cross-prompt AES research, we have chosen to discuss them under this category.

4.2.1 Holistic scoring

4.2.2 Handcrafted feature engineering

Cozma et al. (2018) proposed a method that combines a string kernel and word embedding representation with a Support Vector Machine (SVM) for prompt-specific and cross-prompt AES tasks. The essay script is represented using two approaches: the Histogram Intersection String Kernel (HISK) and the Bag-of-Super-Word-embeddings (BOSWE). The HISK representation captures similarity by considering common n-gram characters, while the BOSWE representation captures semantic similarity by utilizing word2vec embeddings. By combining these two representations and applying an SVM model, a score is predicted for the essay.

This study shows the significance of string kernels in essay representation and overall score prediction. However, string kernels have limitations in capturing semantic information. Although this approach incorporates word embeddings to address this limitation, challenges such as computational complexity, sensitivity to string length, difficulty in capturing structural information, and limited generalization persist, posing challenges for their effective use in AES.

4.2.3 Neural network-based models

In this section, we present studies that employ neural network for cross-domain holistic scoring. In Table 6, we present the summary of these papers with essay representation mechanism and datasets.

Hierarchical architecture: Dong and Zhang (2016) proposed hierarchical holistic scoring architecture for prompt-specific and cross-prompt AES. Their approach introduced word-level and sentence-level convolutional layers, leading to domain adaptation and a significant performance improvement compared to previous models in this category.

Two-stage hybrid neural network: Jin et al. (2018) proposed a two-stage neural network approach, prompt independent stage and prompt-dependent stage. In the first stage (prompt-independent stage), the focus is on selecting representative positive and negative sample data. The selection process of the samples depend on handcrafted features such as word length, number of prepositions, and number of unique words. As it is shown in Fig. 15, RankSVM model is applied over the essay representation to identify essays with low scores and high scores for negative and positive samples to be used in the next stage.

In the second stage (prompt-dependent stage), the main objective is to capture semantic, part-of-speech, and syntactic information. To capture the semantic and the part-of-speech information, the essay is represented using a hierarchical document structure, with word vectors are grouped into sentences, and then sentences are grouped into essays. On the other hand, to capture the syntactic information, essays are transformed into phrases, then sentences, and finally into a complete essay. At each level of the hierarchy, Bi-LSTM is employed to capture contextual dependency. To predict the score, a fully connected linear

Table 6 A summary of neural network-based cross-prompt holistic scoring for AES

	References	Main theme	Essay representation			Datasets
			SW	DM	HF	
The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure(DM), using coherence modeling (CM), and incorporating handcrafted features (HF)	Cozma et al. (2018)	Combines string kernels and super word embeddings with SVM for both prompt-specific and cross-prompt scoring	✓			ASAP-AES
	Dong and Zhang (2016)	A hierarchical CNN architecture for both prompt specific and cross-prompt scoring		✓		ASAP-AES
	Jin et al. (2018)	A two stage approach that selects positive and negative samples and score based on semantic, POS and syntactic information		✓		ASAP-AES
	Ridley et al. (2020)	A convolutional recurrent neural network that takes POS embedding and non prompt-specific features		✓	✓	ASAP-AES
	Cao et al. (2020)	A neural network with domain adversarial training		✓		ASAP-AES

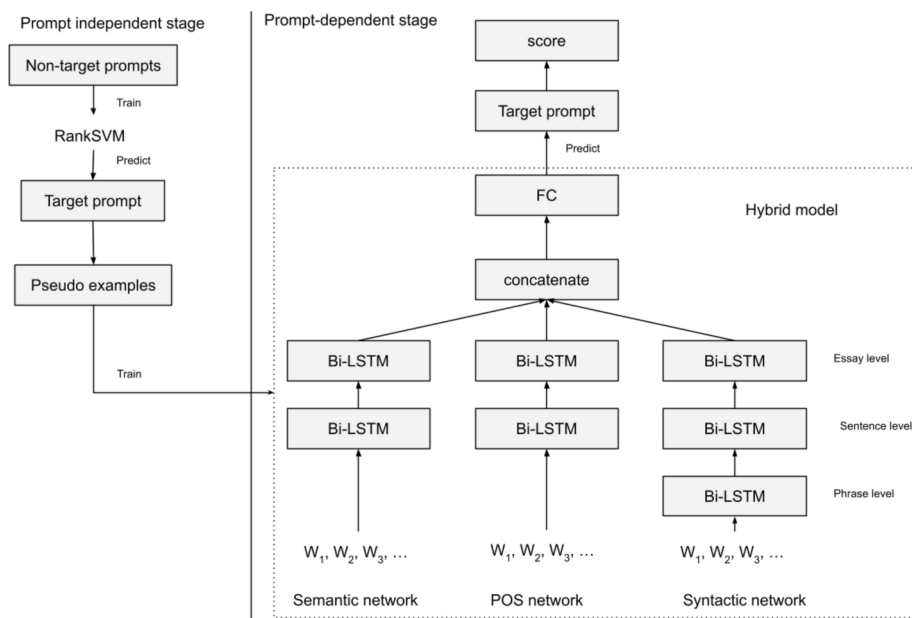


Fig. 15 Two stage deep neural network for prompt independent scoring (Jin et al. 2018)

layer is utilized over the concatenated representation of the semantic, part-of-speech, and syntactic information. This concatenated representation combines the information from all three levels of the hierarchy and is used as input to the scoring model. This two-stage neural network model showed prompt-independent AES model by reducing the impact of contents from the semantic embedding.

POS embedding and non prompt-specific features for generalization: Ridley et al. (2020) proposed a neural network method that employs POS embedding to capture syntactic information from the essays and non prompt-specific features, such as length based, readability, sentiment, text complexity, and text variation. As it is shown in Fig. 16, essay inputs are first converted into part-of-speech (POS) embedding. Subsequently, convolutional recurrent neural networks and pooling methods are employed to extract syntactic features. The final score is computed by concatenating these syntactic features with handcrafted features and passing them through a fully connected linear layer.

The replacement of word embedding by POS embedding showed that a better adaptability level. However, totally ignoring the use of content (semantic) word embedding negatively impacts the performance response-based prompts.

Domain adversarial training: Cao et al. (2020) employ two self-supervised tasks and domain adversarial training to build a cross-domain holistic scoring AES. The proposed method optimizes the training framework with scoring task, sentence reordering task and noise identification task. The scoring task predicts the score of the essay based on the shared representation of the essay. This task utilizes mean squared error to measure how far is the predicted score from the gold reference score. The sentence reordering self-supervised task starts with dividing the essay into four parts and shuffle them into a certain permutation order. The main goal of this task is to predict the order of each sub-part of the essay and

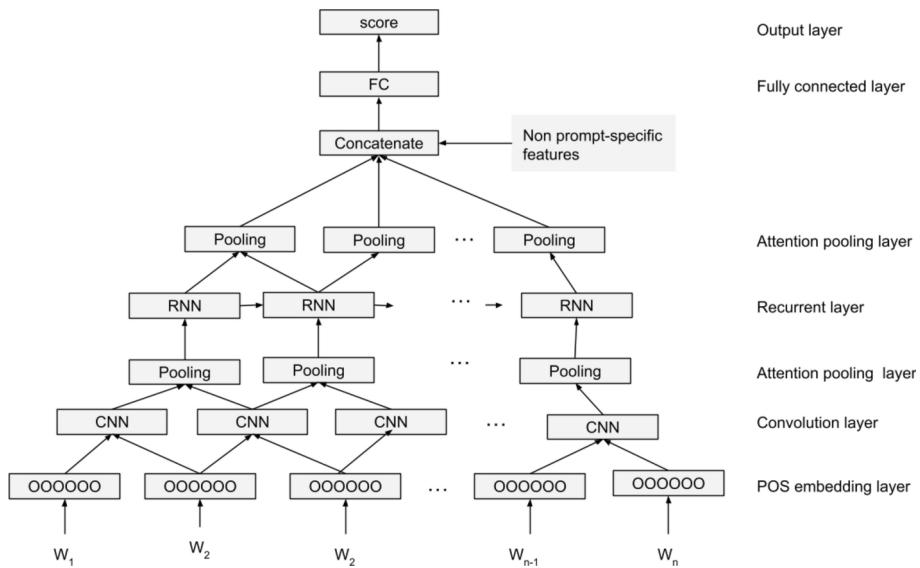


Fig. 16 Architecture of prompt agnostic essay scorer (PAES) (Ridley et al. 2020)

form the original essay. This task uses cross-entropy loss objective function to maintain the original order. The noise identification task also forms a new synthetic data which represents “noise” and “clean” data. The “clean” labeled data is the data directly used as it is from the dataset, whereas the “noise” data is a data altered by inserting, swapping or deleting 10% of the tokens from the source data. The aim of this task is to predict whether the entry is “clean” or “noise” data. Negative log likelihood is used as objective function in this self-supervised task. The aim of employing the self-supervised tasks is to learn a domain independent information.

To further improve the model’s domain adaptability, a perturbation of domain variation that contain information about the target domain is added to the training data. This addition of domain variation information improves the model’s domain adaptation.

4.2.4 Trait-specific AES

4.2.5 Neural network-based models

This section presents studies that employ neural network models for cross-prompt trait specific scoring. We present a summary of the studies in Table 7 with essay representation approaches and employed datasets.

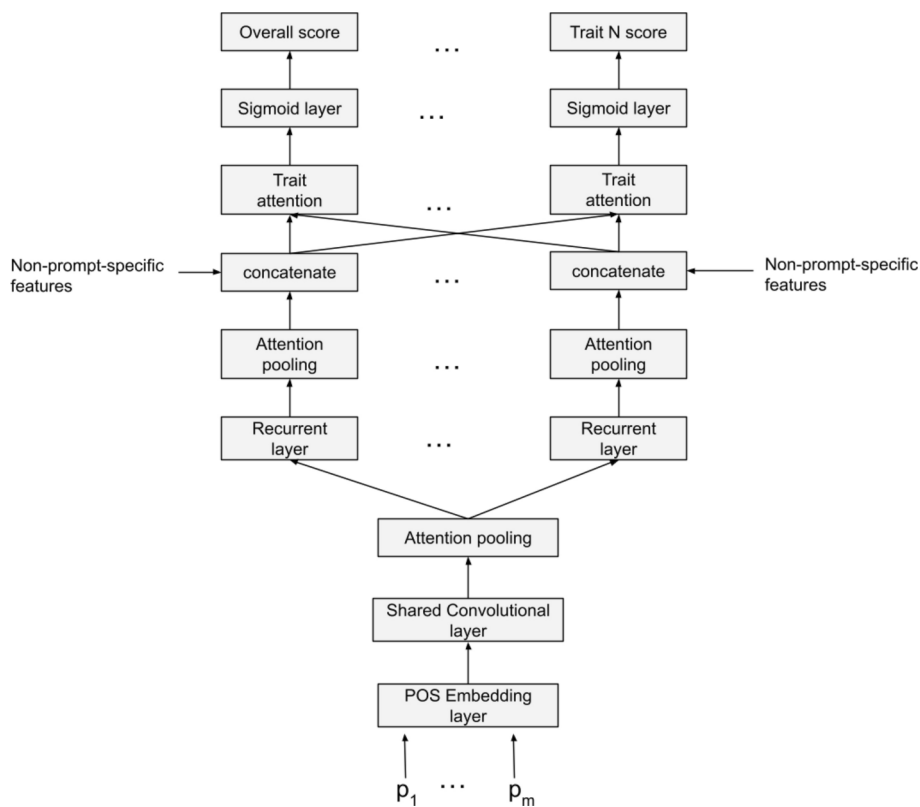
Cross-prompt trait scoring: Ridley et al. (2021) proposed a method to score traits based on cross-prompt setup. In this extension work to Ridley et al. (2020), the method is divided into shared layers and private layers.

As it is shown in Fig. 17, the shared layers represents the common parameters that can be learned from training POS embedding of the essays against the traits and the overall score. The private layers show the training process of each traits and the overall score. The authors

Table 7 A summary of neural network-based cross-prompt trait-specific AES studies

The essay representation column shows that feature representation mechanism, such as representing as sequence of words (SW), hierarchical document structure (DM), using coherence modeling (CM), and incorporating handcrafted features (HF)

References	Main theme	Essay representation			Datasets
		SW	DM	HF	
Ridley et al. (2021)	a multi-task learning (traits) convolutional recurrent neural network that trains on the POS of essay input and non prompt-specific features		✓	✓	ASAP-AES
Chen and Li (2023)	prompt mapping contrastive learning		✓		Not available

**Fig. 17** Architecture of cross-prompt trait scoring (CTS) for AES (Ridley et al. 2021)

compared the performance of cross trait scoring with models primarily design for prompt specific holistic scoring like (Dong et al. 2017) and it is shown that a poor generalization performance in the prompt-specific scoring tasks.

Prompt mapping contrastive learning: Chen and Li (2023) proposed a prompt-mapping contrastive learning strategy for cross-prompt holistic and trait-specific AES tasks called PMAES. The aim of this learning strategy is to capture a consistent representation of essays

in order to obtain more generalizable features from both the source and the target domains. This strategy takes into account the representation of the source prompt and the target prompt while learning shared features. POS embeddings with manual crafted features (Ridley et al. 2021) and hierarchical neural network with attention pooling (Dong et al. 2017) are used to represent the essays. The prompt-mapping contrastive learning strategy learns consistent representation of essays by decreasing the distance between the mapping pairs from source-to-target and target-to-source jointly.

5 Automated feedback generation in AES

Feedback is a crucial element in the evaluation of essay writing. Merely assigning grades to essays does not effectively aid students in enhancing their essay writing abilities. In addition to scoring, guiding students on areas for improvement and how to enhance their writing skills is pivotal in the learning process of essay writing (Patchan et al. 2016). Several studies focus on the generation of feedback in automated essay scoring by identifying the most crucial aspects of the essay that influenced the assigned score. They employ traits such as multi-task scoring to provide feedback based on the relevance level of these traits. Another approach involves utilizing ChatGPT for generating feedback.

5.1 Traits for adaptive feedback

To build upon the research conducted by Taghipour and Ng (2016) in incorporating trait information, Hussein et al. (2020) proposed a model that not only predicts the overall score but also provides trait-specific scores. This approach aims to enhance the performance of the overall score while enabling adaptive feedback based on specific traits. However, a significant challenge arises due to the limited availability of annotated datasets that include trait information.

5.2 Sentence influence estimation task

Woods et al. (2017) utilizes ordinal logistic regression model for essay scoring which attains the state-of-the-art performance on essays with low range of scores while keeping features that are useful for sentence influence estimation task. The authors introduced influence estimation method in order to identify sentences that needs feedback.

5.3 Identifying sentences for feedback

Hellman et al. (2020) proposed an approach called kNN-MIL, which employs k-nearest neighbor (kNN) and multi-instance learning (MIL) models for determining the impact of each sentence in the overall score of the essay. An essay is divided into sentences and each sentence is scored against rubrics to determine its contribution using kNN. This process identifies the sentences that discuss specific topics and rubric designated points across the essay. The aggregated score from the sentences is used to create a document-level holistic score. By framing the AES task as MIL, this work presented sentence annotation mechanism that can enhance the feedback process in building explainable AES model.

5.4 Explainability as a feedback

Kumar and Boulanger (2020) proposed a deep learning approach to automated essay scoring, which provides feedback using explanation. The aim of the proposed approach is to predict the quality of the writing style and expose the decision process that led to these predictions. SHAP is utilized to generate both local and global explanations.

5.5 Rhetorical structure theory parsing for feedback

Fiacco et al. (2022) proposed a neural approach to automated Rhetorical Structure Theory (RST) parsing which can be used to provide structure level feedback by analyzing discourse structure of essays. This work presented an approach that decomposes a written essay and extract hierarchical discourse structure and organization from essays that can be utilized as a feedback for coherent flow of ideas.

5.6 FABRIC

Han et al. (2023a) proposed a *Feedback generation guided with AES By Rubric-based dataset Incorporating ChatGPT* (FABRIC), which is capable of essay scoring and feedback generation. FABRIC combines both essay scoring and feedback generation in the AES task. Fabric has three main components, namely DREsS, CASE, and EssayCoT, which addresses different AES task problems. DREsS is a rubric-based dataset where the traits are content, organization, and language. CASE is a corruption augmentation mechanism to the three rubrics in which AES is trained for essay scoring (Yoo et al. 2024). And, EssayCoT is an essay Chain-of-Thought which is used for feedback generation based on the essay scoring with zero-shot mechanism. BERT is employed for essay representation and ChatGPT for feedback generation.

5.7 Alignment of functional groups with feature groups

Fiacco et al. (2023) proposed a method that can be used to improve the interpretability of Longformer AES models by mapping relationships between feature groups and functional groups. The models in this work are rubric-based model that are trained on organization, main idea, support, and language traits. The mapping of feature groups with functional groups helps to show the decision making process involved in scoring the essays.

6 Dataset

There are several datasets associated with AES tasks, but only a few are free and publicly available. In this section, we will discuss the popular datasets that have been used in recent AES research works. Table 8 shows the widely used datasets in AES. The ASAP and ASAP++ datasets are free and publicly available as well as the widely used datasets in recent years.

Table 8 Widely used datasets in AES researches

Dataset	Writers	Types	Total number of essays
ASAP/ASAP++	Students from grade 7–10	Argumentative, response, narrative	12,900
AI Hub: essay writing evaluation data	Elementary, middle and high school students	argumentative and narrative	50,413
TOEFL11	Non-native speakers	Argumentative and narrative	12,100
ICLE	Non-native speakers	Argumentative and narrative	9529

6.1 ASAP

The ASAP dataset (Hamner et al. 2012) is a popular dataset in AES tasks. It is a free and publicly available dataset published on Kaggle for Automated Student Assessment Prize (ASAP) competition² in 2012. It contains 8 prompts and more than 12,900 essays written by students from grade 7 up to 10. The prompts represent different topics and genres of essays, namely argumentative, response and narrative. Prompt 1 and 2 require opinions of students that can be defined as argumentative essay. Prompts from 3 to 6 ask students to write an essay based on a text provided, and this type of essay is called response essay. The last two prompts, prompt 7 and 8, ask students to write an essay based on a situation which can be defined as a narrative essays.

The ASAP dataset has three prompts with trait features, namely prompt 2, 7, and 8. In prompt 2, the rubric guidelines are categorized into Domain 1 (Writing Applications) and Domain 2 (Language Conventions). Domain 1 is described by ideas and contents, organization, style, and voice. And the score range is 1–6. Domain 2 is about language conventions where the score is based on capitalization conventions, the use of punctuation, spelling, grammar and standard English usage. And the score range for Domain 2 is 1–4. The scores are a holistic scores for each domains. Several research papers considered the score of domain 1 only. Thus, they refer score range for this prompt as 1–6. There is no trait specific score provided in this prompt.

The scoring guideline for Prompt 7 is based on domain 1. The essays are scored against the traits of domain 1. Each traits are scored 0–3 but ideas points are doubled. Therefore, the total score assigned by a rater is 15. Prompt 8 is scored against ideas and contents, organization, voice, conventions, sentence fluency and, word choice traits by two independent raters. In prompt 8, there is a third rater for a resolution. The composite score for this prompt is calculated from ideas, organization, sentence fluency and conventions traits only.

²<https://www.kaggle.com/competitions/asap-aes>

6.2 ASAP++

Mathias and Bhattacharyya (2018) annotated the ASAP-AES dataset with multiple traits by providing a score to each trait. The annotation of traits to the essays depends on the essay type. In the AES dataset, there are three essay types but the argumentative and narrative essays share the same type of trait sets. Thus, the essay sets of traits are divided into two, namely traits of argumentative or persuasive essays and attributes for source-dependent responses.

The traits for the argumentative or persuasive essays are content, organization, word choice, sentence fluency, and conventions whereas for the source-dependent essays the traits are content, prompt adherence, language, and narrativeness.

6.3 AI hub: essay writing evaluation data

AI Hub presented Korean language essay writing evaluation dataset³ that consists 50 topics and 50,413 cases. The 51.5% of them are argumentative essay type where as the 48.5% of them are explanatory essay type. The essays are written by elementary, middle and high school students and are graded by three evaluators into 11 rubric sub categories scores.

6.4 TOEFL11: a corpus of non-native English

TOEFL11⁴ dataset is a corpus collected by Blanchard et al. (2013). It was created by sampling from a set of TOEFL essays written between 2006 and 2007 in response to eight prompts by non-native writers. The dataset contains a total of 12,100 essays, which were graded by two human raters on a 5-point-scale. The scores were later collapsed into a 3-point scale, namely low, medium or high. The length of essays ranges from 2 words to 876 words, with an average length of 348 words.

6.5 ICLE: the international corpus of learner English

The ICLE⁵ is a dataset of essay written by non-native English speakers (Granger et al. 2020). In the latest version of the dataset, ICLEv3, there are 9,529 essays with average length of 605 words. The dataset provides rich metadata on the essay writers and the writing tasks.

³<https://www.aihub.or.kr/aihubdata/data/view.do?currMenu=115&topMenu=100 &aihubDataSe=data &dataSetSn=545>.

⁴https://www.ets.org/research/policy_research_reports/publications/report/2013/jrkv.html.

⁵<https://corpora.uclouvain.be/cecl/icle/home>

7 Evaluation methods

7.1 Quadratic weighted kappa (QWK)

This metrics measures the agreement level between raters. The score ranges from -1 to 1 , where 1 indicates complete agreement, 0 agreement equivalent to chance and -1 complete disagreement.

$$k = 1 - \frac{\sum W_{i,j} O_{i,j}}{\sum W_{i,j} E_{i,j}}$$

where $O_{i,j}$ is a histogram matrix with the number of predicted labels that have a rating of i (actual) that received a predicted value j , $E_{i,j}$ is a histogram matrix of expected ratings calculated as the outer product between the actual rating's histogram vector of ratings and the predicted rating's histogram vector of ratings.

$$W_{i,j} = \frac{(i - j)^2}{(R - 1)^2}$$

where $W_{i,j}$ is a weight matrix that is calculated based on the difference between actual and predicted values, and R is the rating range.

7.2 Mean squared error
Mean squared error (MSE) is the popular loss function used in essay scoring tasks. It is mainly used in the regression aspect of the AES task to optimize the loss. MSE is used to compute how far is the predict value from the reference score.

$$MSE(y, \hat{y}) = \frac{1}{n} \sum (y - \hat{y})^2$$

where n is the number of instances, y is the actual score and $\hat{y}$ is the predicted score.

7.3 Pearson correlation coefficient
Pearson correlation coefficient measures the linear relationship between two raters. It measures the strength and direction of the linear relationship. The score of Pearson's correlation ranges from -1 to 1 .

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum (x_i - \bar{x})^2)} \sqrt{(\sum (y_i - \bar{y})^2)}}$$

where x_i and y_i are actual and predicted scores, and $\bar{x}$ and $\bar{y}$ are the means (averages) of the actual and the predicted scores.

8 Challenges and future research directions in AES
Deep learning models have demonstrated promising results in automating essay grading tasks. However, there are challenges associated with understanding the score prediction process, providing feedback based on scores, and the availability of annotated datasets. In this section, we will delve into these challenges and explore research directions pertaining to AES tasks.

8.1 Availability of limited dataset

The research in AES faces challenges due to the limited availability of annotated datasets. The widely used public AES English dataset, ASAP-AES, comprises fewer than 13,000 essays and only includes two prompts with trait-based scores. This has resulted in a higher number of studies focusing on holistic essay scoring AES models, with less attention given to trait-based AES tasks. However, incorporating traits into AES models can provide explanations and enhance interpretability of AES models. Since collecting essays is both difficult and costly, exploring techniques to pre-train essay models that can enrich essays with traits can be considered as a future research direction.

In addition to the scarcity of annotated data, there is also a limitation in the availability of data in languages other than English. There have been few studies that have utilized datasets in languages other than English (Hirao et al. 2020; Song et al. 2020b; He et al. 2022; Song et al. 2020b, b). Consequently, the availability of more datasets and AES models trained in different languages can help bridge the gap in democratizing AES systems for underrepresented communities and societies.

8.2 Automated feedback generation in AES

Although feedback is an essential and integral component of essay writing in educational settings, its study within the context of deep learning-based AES models is relatively limited. Identifying the essay components that contribute to the overall score can be a valuable task as it enables writers to identify the weak areas within their essays and improve upon them. Therefore, investigating neural feedback-based methods would be a promising research direction to enhance the effectiveness of AES models and provide more insightful feedback to writers.

8.3 Interpretability of AES models

This survey paper discusses various deep learning-based AES models that excel at identifying complex patterns within essays. However, understanding the internal workings of these models and their relationship with predicted scores can be challenging. Explainable AI (XAI) is a research area that focuses on revealing the features and patterns used to make decisions, aiming to build trust in AI models. Kumar and Boulanger (2020) emphasize the significance of explainable AES models for pedagogy. Exploring XAI in general is an intriguing research topic, and its application to AES models would greatly facilitate the uncovering of features utilized in the scoring process.

8.4 Creative writing in essays

Creativity in essay writing is a valuable skill for students as it allows them to express their opinions in a unique and engaging manner. Assessing the engagement of creative writing and incorporating it into the essay score can greatly enhance overall evaluations in classroom settings.

However, measuring creativity in essay writing evaluation is a challenging task. The evaluation metric for creativity encompasses attributes such as fluency, flexibility, original-

ity, elaboration, and appropriateness (Lee et al. 2023), which are difficult to quantify. Only a few studies have explored creative writing in AES tasks, particularly utilizing deep learning methods (Liang et al. 2021; Zhang et al. 2022; Lee et al. 2023). Hence, we propose that investigating creativity within the context of AES would be an intriguing research direction.

8.5 Adversarial crafted essays

One of the challenges in AES is effectively capturing adversarially crafted essay inputs, such as those with grammatically correct sentences but poor coherence, or even AI generated essays. Only few studies have addressed these issues by employing coherence-based models (Farag et al. 2018; Liu et al. 2019) and utilizing large language models (Peng et al. 2024). Therefore, investigating adversarial scenarios in AES presents an interesting research topic that can contribute to enhancing the robustness and reliability of AES systems.

8.6 Prompt independent models

In addition to the challenge of limited annotated essay datasets, the process of creating labeled datasets for every domain of essays is impractical and costly. The absence of labeled target data across various domains highlights the need for research in prompt-independent models. Therefore, exploring models for cross-prompt AES becomes an intriguing and worthwhile research topic.

9 Conclusion

In this survey paper, we have presented a comprehensive review of existing deep learning-based AES models. Our analysis delved into essay representation techniques, and various essay scoring models. Recognizing the importance of feedback in writing assessments, we also explored studies that focus on feedback identification and generation. Additionally, we discussed widely used AES datasets and evaluation metrics. It is worth noting that although many of the datasets used in AES research are not publicly available, it is possible to obtain access by establishing proper communication with the hosting organizations. However, there are free and publicly available datasets, such as ASAP and ASAP++, which lack feedback associated with the scores.

This work presents the challenges and research directions in AES. We have highlighted the limited research on feedback generation in essay scoring. We hypothesize that the gap in feedback generation for AES may be attributed to the scarcity of annotated datasets that include feedback and traits. However, recent studies (Han et al. 2023a) have demonstrated that the augmentation of large language models can alleviate the challenge of limited feedback in datasets.

In summary, the study of AES tasks holds great importance due to its applications in education and other writing support systems. Therefore, it is crucial to investigate efficient methods for information representation that can be effectively utilized for both scoring and feedback purposes. Pursuing this research direction can contribute to the development of robust AES models.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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References

- Alikaniotis D, Yannakoudakis H, Rei M (2016) Automatic text scoring using neural networks. arXiv preprint [arXiv:1606.04289](https://arxiv.org/abs/1606.04289)
- Bahdanau D, Cho K, Bengio Y (2014) Neural machine translation by jointly learning to align and translate. arXiv preprint [arXiv:1409.0473](https://arxiv.org/abs/1409.0473)
- Blanchard D, Tetreault J, Higgins D et al (2013) (2013) Toefl11: a corpus of non-native english. ETS Res Rep Ser 2:i–15
- Bojanowski P, Grave E, Joulin A, et al (2016) Enriching word vectors with subword information. arXiv preprint [arXiv:1607.04606](https://arxiv.org/abs/1607.04606)
- Cao Y, Jin H, Wan X, et al (2020) Domain-adaptive neural automated essay scoring. In: Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval, pp 1011–1020
- Carlile W, Gurrupadi N, Ke Z, Ng V (2018) Give me more feedback: Annotating argument persuasiveness and related attributes in student essays. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp 621–631
- Chassab RH, Zakaria LQ, Tiun S (2021) Automatic essay scoring: A review on the feature analysis techniques. International Journal of Advanced Computer Science and Applications 12(10)
- Chen Y, Li X (2023) Pmaes: prompt-mapping contrastive learning for cross-prompt automated essay scoring. In: Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers), pp 1489–1503
- Cho B, Jang Y, Yoon J (2023) Rubric-specific approach to automated essay scoring with augmentation training. arXiv preprint [arXiv:2309.02740](https://arxiv.org/abs/2309.02740)
- Cozma M, Butnaru AM, Ionescu RT (2018) Automated essay scoring with string kernels and word embeddings. arXiv preprint [arXiv:1804.07954](https://arxiv.org/abs/1804.07954)
- Dasgupta T, Naskar A, Dey L, et al (2018) Augmenting textual qualitative features in deep convolution recurrent neural network for automatic essay scoring. In: Proceedings of the 5th workshop on natural language processing techniques for educational applications, pp 93–102
- Devlin J, Chang MW, Lee K, et al (2018) Bert: pre-training of deep bidirectional transformers for language understanding. arXiv preprint [arXiv:1810.04805](https://arxiv.org/abs/1810.04805)

- Do H, Kim Y, Lee GG (2023) Prompt-and trait relation-aware cross-prompt essay trait scoring. arXiv preprint [arXiv:2305.16826](https://arxiv.org/abs/2305.16826)
- Dong F, Zhang Y (2016) Automatic features for essay scoring—an empirical study. In: Proceedings of the 2016 conference on empirical methods in natural language processing, pp 1072–1077
- Dong F, Zhang Y, Yang J (2017) Attention-based recurrent convolutional neural network for automatic essay scoring. In: Proceedings of the 21st conference on computational natural language learning (CoNLL 2017), pp 153–162
- Farag Y, Yannakoudakis H, Briscoe T (2018) Neural automated essay scoring and coherence modeling for adversarially crafted input. arXiv preprint [arXiv:1804.06898](https://arxiv.org/abs/1804.06898)
- Fiacco J, Adamson D, Ros C (2023) Towards extracting and understanding the implicit rubrics of transformer based automatic essay scoring models. In: Proceedings of the 18th workshop on innovative use of NLP for building educational applications (BEA 2023), pp 232–241
- Fiacco J, Jiang S, Adamson D, et al (2022) Toward automatic discourse parsing of student writing motivated by neural interpretation. In: Proceedings of the 17th workshop on innovative use of NLP for building educational applications (BEA 2022), pp 204–215
- Galhardi LB, Brancher JD (2018) Machine learning approach for automatic short answer grading: a systematic review. In: Advances in artificial intelligence-IBERAMIA 2018: 16th Ibero-American conference on AI, Trujillo, Peru, November 13–16, 2018, Proceedings 16. Springer, pp 380–391
- Granger S, Dupont M, Meunier F, et al (2020) The international corpus of learner english (version 3). Presses Universitaires de Louvain
- Haller S, Aldea A, Seifert C, et al (2022) Survey on automated short answer grading with deep learning: from word embeddings to transformers. arXiv preprint [arXiv:2204.03503](https://arxiv.org/abs/2204.03503)
- Hamner B, Morgan J, Lynnvandev M, et al (2012) The Hewlett foundation: automated essay scoring. Kaggle
- Han S, Yoo DG, On BW et al (2023) Empirical study on the loss functions of contrastive learning-based multi-scale bert model for automated essay scoring. J Korean Inst Inf Technol 21(9):51–63
- Han J, Yoo H, Myung J, et al (2023a) Fabric: automated scoring and feedback generation for essays. arXiv preprint [arXiv:2310.05191](https://arxiv.org/abs/2310.05191)
- Havnes A, Smith K, Dyshe O et al (2012) Formative assessment and feedback: Making learning visible. Stud Educ Eval 38(1):21–27
- He Y, Jiang F, Chu X, et al (2022) Automated chinese essay scoring from multiple traits. In: Proceedings of the 29th international conference on computational linguistics, pp 3007–3016
- Hellman S, Murray WR, Wiemerslage A, et al (2020) Multiple instance learning for content feedback localization without annotation. In: Proceedings of the 15th workshop on innovative use of NLP for building educational applications, pp 30–40
- Hicke Y, Tian T, Jha K, et al (2023) Automated essay scoring in argumentative writing: Debteachingassistant. arXiv preprint [arXiv:2307.04276](https://arxiv.org/abs/2307.04276)
- Hirao R, Arai M, Shimanaka H, et al (2020) Automated essay scoring system for nonnative japanese learners. In: Proceedings of the 12th language resources and evaluation conference, pp 1250–1257
- Hoblos J (2020) Experimenting with latent semantic analysis and latent dirichlet allocation on automated essay grading. In: 2020 7th international conference on social networks analysis, management and security (SNAMS). IEEE, pp 1–7
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780
- Hussein MA, Hassan HA, Nassef M (2020) A trait-based deep learning automated essay scoring system with adaptive feedback. Int J Adv Comput Sci Appl 11(5)
- Hussein MA, Hassan H, Nassef M (2019) Automated language essay scoring systems: a literature review. PeerJ Comput Sci 5:e208
- Jiang Z, Gao T, Yin Y, et al (2023) Improving domain generalization for prompt-aware essay scoring via disentangled representation learning. In: Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers), pp 12456–12470
- Jin C, He B, Hui K, et al (2018) Tdnn: a two-stage deep neural network for prompt-independent automated essay scoring. In: Proceedings of the 56th annual meeting of the association for computational linguistics (volume 1: long papers), pp 1088–1097
- Jong YJ, Kim YJ, Ri OC (2023) Review of feedback in automated essay scoring. [arXiv:2307.05553](https://arxiv.org/abs/2307.05553)
- Ke Z, Ng V (2019) Automated essay scoring: a survey of the state of the art. In: IJCAI, pp 6300–6308
- Klebanov BB, Madnani N (2020) Automated evaluation of writing—50 years and counting. In: Proceedings of the 58th annual meeting of the association for computational linguistics, pp 7796–7810
- Kumar V, Boulanger D (2020) Explainable automated essay scoring: deep learning really has pedagogical value. In: Frontiers in education. Frontiers Media SA, p 572367
- Kumar R, Mathias S, Saha S, et al (2021) Many hands make light work: using essay traits to automatically score essays. arXiv preprint [arXiv:2102.00781](https://arxiv.org/abs/2102.00781)

- Lagakis P, Demetriadis S (2021) Automated essay scoring: a review of the field. In: 2021 International conference on computer, information and telecommunication systems (CITS). IEEE, pp 1–6
- LeCun Y, Bengio Y, Hinton G (2015) Deep learning. *Nature* 521(7553):436–444
- Lee Y, Kim D, On BW et al (2023) A comparative analysis of the effectiveness of rare tokens on creative expression using rambert. *Find Assoc Comput Linguist: ACL 2023*:10063–10077
- Li F, Xi X, Cui Z et al (2023) Automatic essay scoring method based on multi-scale features. *Appl Sci* 13(11):6775
- Liang G, On BW, Jeong D et al (2018) Automated essay scoring: a siamese bidirectional lstm neural network architecture. *Symmetry* 10(12):682
- Liang G, On BW, Jeong D et al (2021) A text gan framework for creative essay recommendation. *Knowledge-Based Syst* 232:107501
- Li X, Chen M, Nie J, et al (2018) Coherence-based automated essay scoring using self-attention. In: Chinese Computational Linguistics and Natural Language Processing Based on Naturally Annotated Big Data: 17th China National Conference, CCL 2018, and 6th International Symposium, NLP-NABD 2018, Changsha, China, October 19–21, 2018, Proceedings 17. Springer, pp 386–397
- Liu J, Xu Y, Zhu Y (2019) Automated essay scoring based on two-stage learning. *arXiv preprint arXiv:1901.07744*
- Mathias S, Bhattacharyya P (2018) Asap++: enriching the asap automated essay grading dataset with essay attribute scores. In: Proceedings of the 11th international conference on language resources and evaluation (LREC 2018)
- Mayfield E, Black AW (2020) Should you fine-tune bert for automated essay scoring? In: Proceedings of the 15th workshop on innovative use of NLP for building educational applications, pp 151–162
- Mesgar M, Strube M (2018) A neural local coherence model for text quality assessment. In: Proceedings of the 2018 conference on empirical methods in natural language processing, pp 4328–4339
- Mikolov T, Chen K, Corrado G, et al (2013) Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*
- Mizumoto A, Eguchi M (2023) Exploring the potential of using an ai language model for automated essay scoring. *Res Methods Appl Linguist* 2(2):100050
- Nadeem F, Nguyen H, Liu Y, et al (2019) Automated essay scoring with discourse-aware neural models. In: Proceedings of the 14th workshop on innovative use of NLP for building educational applications, pp 484–493
- Owen L (2016) The impact of feedback as formative assessment on student performance. *Int J Teach Learn High Educ* 28(2):168–175
- Patchan MM, Schunn CD, Correnti RJ (2016) The nature of feedback: How peer feedback features affect students' implementation rate and quality of revisions. *J Educ Psychol* 108(8):1098
- Peng X, Zhou Y, He B, et al (2024) Hidding the ghostwriters: an adversarial evaluation of ai-generated student essay detection. *arXiv preprint arXiv:2402.00412*
- Pennington J, Socher R, Manning CD (2014) Glove: global vectors for word representation. In: Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP), pp 1532–1543
- Persing I, Ng V (2015) Modeling argument strength in student essays. In: Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), pp 543–552
- Ramesh D, Sanampudi SK (2022) An automated essay scoring systems: a systematic literature review. *Artif Intell Rev* 55(3):2495–2527
- Ridley R, He L, Dai Xy, et al (2021) Automated cross-prompt scoring of essay traits. In: Proceedings of the AAAI conference on artificial intelligence, pp 13745–13753
- Ridley R, He L, Dai X, et al (2020) Prompt agnostic essay scorer: a domain generalization approach to cross-prompt automated essay scoring. *arXiv preprint arXiv:2008.01441*
- Rodriguez PU, Jafari A, Ormerod CM (2019) Language models and automated essay scoring. *arXiv preprint arXiv:1909.09482*
- Smolentzov A (2013) Automated essay scoring: Scoring essays in Swedish.
- Song W, Song Z, Liu L, et al (2020a) Hierarchical multi-task learning for organization evaluation of argumentative student essays. In: *IJCAI*, pp 3875–3881
- Song W, Zhang K, Fu R, et al (2020b) Multi-stage pre-training for automated chinese essay scoring. In: Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), pp 6723–6733
- Stab C, Gurevych I (2014) Annotating argument components and relations in persuasive essays. In: Proceedings of COLING 2014, the 25th international conference on computational linguistics: Technical papers, pp 1501–1510
- Stab C, Gurevych I (2016) Recognizing the absence of opposing arguments in persuasive essays. In: Proceedings of the Third Workshop on Argument Mining (ArgMining2016), pp 113–118

- Stab C, Gurevych I (2017) Parsing argumentation structures in persuasive essays. *Comput Linguist* 43(3):619–659
- Taghipour K, Ng HT (2016) A neural approach to automated essay scoring. In: *Proceedings of the 2016 conference on empirical methods in natural language processing*, pp 1882–1891
- Tay Y, Phan M, Tuan LA, et al (2018) Skipflow: incorporating neural coherence features for end-to-end automatic text scoring. In: *Proceedings of the AAAI conference on artificial intelligence*
- Uto M (2021) A review of deep-neural automated essay scoring models. *Behaviormetrika* 48(2):459–484
- Uto M, Aomi I, Tsutsumi E, Ueno M (2023) Integration of prediction scores from various automated essay scoring models using item response theory. *IEEE Transact Learn Technol* 16(6):983–1000
- Uto M, Okano M (2020) Robust neural automated essay scoring using item response theory. In: *Artificial intelligence in education: 21st international conference, AIED 2020, Ifrane, Morocco, July 6–10, 2020, proceedings, part I 21*. Springer, pp 549–561
- Uto M, Xie Y, Ueno M (2020) Neural automated essay scoring incorporating handcrafted features. In: *Proceedings of the 28th international conference on computational linguistics*, pp 6077–6088
- Wang X, Lee Y, Park J (2022a) Automated evaluation for student argumentative writing: a survey. *arXiv preprint arXiv:2205.04083*
- Wang Y, Wang C, Li R, et al (2022b) On the use of bert for automated essay scoring: joint learning of multi-scale essay representation. *arXiv preprint arXiv:2205.03835*
- Wang Y, Wei Z, Zhou Y, et al (2018) Automatic essay scoring incorporating rating schema via reinforcement learning. In: *Proceedings of the 2018 conference on empirical methods in natural language processing*, pp 791–797
- Williamson DM, Bejar II, Hone AS (1999) ‘Mental model’ comparison of automated and human scoring. *J Educ Meas* 36(2):158–184
- Woods B, Adamson D, Miel S, et al (2017) Formative essay feedback using predictive scoring models. In: *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining*, pp 2071–2080
- Xie J, Cai K, Kong L, et al (2022) Automated essay scoring via pairwise contrastive regression. In: *Proceedings of the 29th international conference on computational linguistics*, pp 2724–2733
- Xue J, Tang X, Zheng L (2021) A hierarchical bert-based transfer learning approach for multi-dimensional essay scoring. *IEEE Access* 9:125403–125415
- Yancey KP, Laflair G, Verardi A, et al (2023) Rating short I2 essays on the cefr scale with gpt-4. In: *Proceedings of the 18th workshop on innovative use of NLP for building educational applications (BEA 2023)*, pp 576–584
- Yang R, Cao J, Wen Z et al (2020) Enhancing automated essay scoring performance via fine-tuning pre-trained language models with combination of regression and ranking. *Find Assoc Comput Linguist: EMNLP 2020*:1560–1569
- Yoo H, Han J, Ahn SY, et al (2024) Dress: dataset for rubric-based essay scoring on efl writing. *arXiv preprint arXiv:2402.16733*
- Zhang S (2021) Review of automated writing evaluation systems. *J China Comput-Assist Lang Learn* 1(1):170–176
- Zhang D, Zhang M, Peng C et al (2022) Expressing metaphorically, writing creatively: metaphor identification for creativity assessment in writing. *Companion Proc Web Confer 2022*:1198–1205

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